



Organosilver(I) and organozinc(II) catalysed synthesis of quaterphenyls – Experimental and theoretical treatment



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ABSTRACT

An essential part of molecular scaffolds of organosilver(I) catalysing reactions of C–H bond activation and C–X (X = C or O) bond formation of polysubstituted benzoic acids in polar medium represent biphenyls, 6H-benzo-[c]-chromene[6]-one, chromeno[5,4,3-cde]chromene-5,10-dione, and 4,10-dioxo-pyrene-5,9-diones. A second strand of catalytic activity is in promoting synthesis of substituted quaterphenyls. This paper is an extend of the systematic examining of catalytic activity of Ag^I-containing intermediates of substituted benzoic acids, gaining insights by synthesis of oligomers, studying benzene-1,2,4-tricarboxylic acid (124BA) and benzene-1,3,5-tricarboxylic acid (135BA). Arguably, the optimization of the latter reactions need further effort in order to achieve the standard characteristics allowing their effective implementation as routine routs in the field of organic synthesis. The purpose of this part of the research is mass spectrometric (MS) quantitative kinetic and thermodynamic treatment of collision induced dissociation (CID) reactions comparing experimental reaction kinetics and energetics with quantum chemical modelling of MS fragment processes. Analyses - experimental and theoretical - include deuterium labelled oligomers as the most useful technique studying MS reaction mechanisms. More specifically we treat eight substituted quaterphenyls both experimentally and theoretically along with a set of one hundred twenty 3D molecular models of MS fragment species. Importantly, the detail understanding of the molecular mechanistic aspects of the reactions causing for obtaining of high functionalized oligomers allows an efficient optimization of synthetic roads studying polyfunctional derivatives. This knowledge is the key to understanding why the obtaining of the oligomer derivatives depends on the positional isomerism of the starting reagents as clearly evidenced in this study. In this context the study is relevant to the fields of organometallic synthesis and catalysis as well as the polymers' chemistry.

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1. Introduction

In recent decades the number of papers dealing with organometallic catalysing C–H activation and C–C/C–X (X = C, N, O, S, P, etc.) bond formation has increased considerably. The contributions to the fields of the organic synthesis and catalysis have become even significantly more numerous than studies devoted to many other known organic reactions. A reason for this growing interest to the organometallic catalysing C–H activation is the fact that there has capability for synthesis of very complex from a structural

chemistry point of view molecular scaffolds using simple starting organic reagents [1]. There can be produced key molecular fragments found in diverse number of naturally occurring products, the largest part of which exhibit unique biological activity [1]. In addition, the discussed reaction is broadly used, amongst others, in the synthesis of heterocyclic, which are the major templates to a significant part of the implemented in the clinical practice pharmaceuticals as well as many well-known families of natural products, for instance, alkaloids, porphyrins, penicillin, etc. [11]. Moreover the discussed reactions exhibit great synthetic efficiency under mild conditions, thus making C–H activation/C–X bond formation processes key ones among many other synthetic roads, thus leading to their employment in the green chemistry research and development of the efficient reactions of C–X couplings (X = C, N, O, S, P, etc.) in water [1i,m,r,s]. To a great extend the latter facts determine

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the known systematic efforts in the field of organometallic synthesis *via* transition metal (TM) catalysed C–H activation, because of the reactions yield to products in good-to-excellent yields under mild experimental conditions, involving economical and straightforward pathways, and in parallel tolerating, a large scale of chemical substitution [1f]. On the other hand, polyfunctional carboxylic acids have been regarded as most promising and economical organic reagents, amongst others, for C–C coupling reactions and C–H activation, due to their great capability for catalytic decarboxylation [1b,c,g,h,j,k,w]. Explicitly, it should be mentioned that benzoic acids (BAs) are in particular very attractive templates due to their ability to participate in step-economic organic synthetic roads as a result from a great chemical reactivity [1g,h,j]. In addition to the fact that substituted BAs are broadly found as molecular scaffolds on many naturally occurring products and synthetic medications exhibiting a broad spectrum of biological function [1h]. Among known transition metal complexes, largest part contributions to the organometallic C–H activation/C–X couplings have been associated with Pd^{II}-containing intermediate, which, in general, have proven catalytic activity tolerating both C(sp³)-H and C(sp²)-H activation. In this context there have been reported Pd^{II}-catalysed synthesis of amides, substituted pyridines, a large scale of other N- and O-containing derivatives [2]. However, looking on the relevant literature there can be seen that the largest part of Pd^{II}-catalysing reactions have been carried out in the presence of so-called additives, representing complex salts of TM ions, in particular underlying Ag^I-ion containing complexes. The role of these additives to the catalytic efficiency has been regarded as a second one [1,2]. However, there has been found that within the framework of many organopalladium(II) catalysed C–H activation reactions, the participating Ag^I-intermediates cannot be replaced with other TM ion containing derivatives. It has been shown in a large set of synthetic works that the catalytic efficiency of Pd^{II}-catalysed reactions of C–H activation/C–N bond formation is increased in 70% using Ag^I-acetate comparing with the yields of the processes in presence of only Pd^{II}-catalysing agent [3]. Moreover an improvement of the reactions yields has been achieved even replacing Pd^{II}- salt with corresponding Ag^I-acetate using additionally Cs-ion containing salt as additive (Yields up to 97% have been reported) [3]. As has been comprehensively discussed in Ref. [4] these results arise the question about the role of Ag^I-ions in C–H activation/C–X coupling mechanisms [4]. Due to the, in general, a lack of systematic fundamental studies devoted to insights of competitive catalytic activity of Pd^{II} and Ag^I the role of Ag^I-ions in organopalladium(II) catalysing processes as well as the role of additive TM ion containing salts to corresponding Pd^{II}-catalyst, in a general context, remain far from well understood. And the discussion consist of few more recent contributions to this topic [1c,3,4]. Furthermore the Ag^I-salts can play a role of an additive, co-catalyst and/or catalyst [1c,t,u,v,3,4]. It appears from the corresponding small amount of available references of the latter topic that the competitive activity of the main catalysts and corresponding other TM ion containing additives are more complex and its understanding requires systematic effort devoted to the fundamental mechanistic molecular level aspects of the catalytic activity both employing theoretical and experimental methods [1c,3,4]. In order to get a detail picture of this topic more recently there has been comprehensively studied Pd^{II}-catalysed C–H activation of N-arylbenzamides and corresponding C–N bond formation [3]. Explicitly has been highlighted, evaluating the energetics in terms of 'free Gibbs energy' of the catalytic reactions, that the presence of Ag^I-acetate causes for decreasing in the activation energy of the pathway [3]. However, it should be pointed out that a more general picture of the mechanistic aspects needs a comparative analysis including various additive salts if Ag^I-ions, because of as it has

been discussed [3] the stoichiometry composition of the geometry of interacting ensembles at the transition states varies. So, an adequate description of the organometallic catalysed processes in a general context needs a comparative theoretical analysis and experimental designs accounting for different molecular factors and experimental conditions. Most importantly, however, are the data revealing a direct correlation between theoretical quantum chemical modelling of the catalytic activity of C–H activation/C–X bond formation reactions (X = C, N, O, S, P, etc.) and experimental synthetic data about the thermochemistry of the reactions (See details in Refs. [3] and [4]). In this context our work presented herein as well as the study on the major products of interaction of Ag^I- and Zn^{II}-catalysing processes of polyfunctional BAs [1c] are very important reports gaining insights *into* these catalytic reactions, thus allowing further optimization of the synthetic roads according the molecular factors determining the catalytic activity in the TS together with the results from the effect of the environmental parameters on the synthetic efficiency. Because of our theoretical and experimental designs include a comparative mass spectrometric and theoretical quantum chemical treatment of the reaction mechanisms, thermodynamics and kinetics. Of course, it would be quite premature to claim that our examining provides unique understanding of the Ag^I-catalysing reactions yielding to substituted biphenyl, 6H-benzo-[c]-chromene[6]-one, chromeno [5,4,3-cde]chromene-5,10-dione, and 4,10-dioxo-pyrene-5,9-dione [1c] or substituted quaterphenyls and their derivatives (this paper) because of as it has been illustrated in Ref. [3] further insights can be gained studying a large scale of: (i) differently substituted benzoic acids, in order to account for the effect of the type of the substituents; (ii) different type Ag^I-complexes as catalysts, in order to account for the effect of the geometry of the chromophore on the energetics at TS; (iii) additional additives including simple alkali salts, which themselves indicate to study the reaction under the effect of the ionic strength; (iv) employment of various solvent types depending on their proton donating and accepting ability as well as polarity; and (v) other experimental factors such as pH, temperature, pressure, etc. However it is important to reiterate that such as research effort is highly justified. Because of, if one was to look at the large body of references, one would find that the employment of only Ag^I-containing intermediates has yielded to numerous promising results about the catalytic efficiency of Ag^I-containing intermediates even reducing the literature research to the discussed here organic reactions [1c,d,g,h,j,3–5]. Moreover the new results reported in this study additionally underline the great prospective of employment of Ag^I-ion containing catalysts causing for isolation of oligomer highly functionalized derivatives from simple starting organics under mild experimental conditions. The work presented herein is triggered to a significantly great extend namely by the obtained surprising high catalytic regioselectivity of Ag^I-containing organometallics in addition to the obtained highly functionalized oligomers of polyfunctional BAs. This, in fact, represents a significant innovation in the field of C–H activation/C–C bond formation reactions known to this date and reported first in the literature. In other words, the catalytic efficiency of complexes of Ag^I-ion is far from well understood. In parallel, known reports provide evidences about a prominent and very prospective for the fields of the organic synthesis and catalysis results, from the perspective of the structural chemistry towards effort associated with optimization of synthetic roads yielding to highly functionalize molecular scaffolds of organic compounds. The major reason for this statement is based on the fact that, in general, the coordination chemistry of Ag^I-ion exhibits a diversity of unusual metal chromophores and structures with, in parallel, large number of ligands [6]. In addition, the chemistry of the silver is associated with various oxidation states (Ag^I, Ag^{II}, Ag^{III} and Ag⁰)

which within the framework of several possible redox reactions could affect additionally the thermodynamic stability *into* direction 'obtaining of a major product'. We would like to mention that the latter property of the silver has been described, in general, as a disadvantage comparing with the chemical reactivity of TM ions with stable oxidation state. However, a detail study on the thermodynamic stability of the organometallic intermediates in the TS indicates that very frequently their stability is governed by catalytic redox reaction [1c]. Because of a detail study of plenty of data devoted to Pd^{II}–catalysed C–H activation indicates that depending on the reaction mechanism a change from Pd⁰/Pd^{II} to Pd^{II}/Pd^{IV} can be found [1g,h,5ab,ac]. Or from a perspective of organosilver(I) catalysis the unstable oxidation state of the silver could be regarded as a rather advantage, than a disadvantage, comparing with other elements from the TM series. Given that complexes of Ag^I–ion are excellent templates for the organometallic catalysts applicable to a large number of reactions depending on the complexity of the mechanisms and the molecular scaffolds of the products.

To address both the organometallic catalysed synthetic needs and theoretical problematic of the molecular level mechanistic aspects of the reactions we utilize complementary the experimental high resolution electrospray ionization mass spectrometry and computational quantum chemistry. The great applicability of the mass spectrometry, in a general context, to study organometallic catalysed reactions derives from its set of instrumental advantages, for instance, high resolving power allowing identification and determination of analytes in complex multicomponent mixtures; low concentration limits of detection and quantification; capability to provide qualitative, quantitative and structural information together with thermodynamic and kinetics ones. Thus there advantages and superior instrumental characteristics of the method take irreplaceable place in the field of organometallic synthesis and catalysis. Or the instrumental characteristics of the mass spectrometry are the dominant motives for its utilization in this study. Its complementary application with the powerful quantum chemical methods on the other hand allow answering questions associated with insights of the reaction mechanisms. Given that, such as studies provide crucial information about the relationship molecular structure and factors governing the chemical reactivity and the reaction efficiency in both thermodynamic and kinetic terms. Therefore, our analyses presented in this paper and interpretation of both theoretical and experimental data provide comprehensive knowledge about the molecular and environmental factors governing the organosilver(I) catalysed synthesis of quaterphenyls using simple and cheap starting reagents, thus allowing an optimization of the reaction efficiency and its further routine implementation in the field of organic synthesis.

2. Experimental

2.1. Physical measurements

High performance liquid chromatography (HPLC) ESI–MS/MS measurements were carried out using TSQ 7000 instrument (Thermo Fisher Inc., Rockville, MD, USA). A triple quadrupole mass spectrometer (TSQ 7000 Thermo Electron, Dreieich, Germany) equipped with an ESI 2 source was utilized (Table S1). A standard LTQ Orbitrap XL (Thermo Fisher Inc.) was employed as well. The ImageQuest 1.0.1 program package was applied to data processing (Thermo Fisher Inc.). The quantification was carried out by trap, linear ion trap and orbitrap mass detectors for time (*t*) = 7–25 min. Selected reaction monitoring method was introduced. The raw data were saved as individual files. The MS spectra of the samples were obtained as average MS patterns between *t* = 0.2–0.6 min after infusion processes in the source. They were measured in triplicate,

in order to account for instrumental repeatability and its influence on analytical signal. In order to avoid a carryover effect of ion source, injection time (*t*) was 6 min. The MS spectra resolution is 95 501 values. The connection with ionization ion source and auto-sampler was made by a stainless steel capillary. Its length was 2 m. The internal diameter was 0.12 mm. The T of column thermostat was 35 °C. The ESI–MS operation conditions were: drying gas T = 350 °C; drying gas flow 8 L min^{−1}; pressure (T) of nebulizing gas P = 40 psi; potential applied to extraction capillary 4000 V. The measurements were performed *via* direct infusion of analyte solutions in ion source, operating in ESI(+) and ESI(−) modes. The analyte flows were by CH₃CN (Gradient grade, Merck, Darmstadt, Germany) and 0.1% aqueous solution of HCOOH (*v/v*) (Eluent additive for liquid chromatography grade, Sigma-Aldrich, Germany) in a volumetric ratio *v/v* = 1:1. The flow rate was 0.5 μL min^{−1}. The syringe diameter is 2.3 mm. The injection volume is shown in Table S1. The analyte solutions were prepared by adding of 2 mL CH₃OH (Uvasol®, Merck, Darmstadt, Germany) to 100 mg of solid peptide, preliminary placed in vials, which were sealed. They were additionally vortexes for *t* = 1 min at 2500 rpm. The solutions were kept at T = 25 °C. The pH and ionic strengths were optimized adding corresponding amounts (μL) of aqueous solutions of NaOH or NaCl, preliminary correcting the volume of mobile phase composition of the vials. In order to account for any variability induced due to sample preparation steps, there were conducted independent set sample preparation procedures at a time duration scale from first experiment of 1 and 2 months, respectively. The chromatographic analysis about purity of peptides before their MS determination, was carried out by GynkoteK (Germering, Germany) HPLC instrument, equipped with a preparative Kromasil 100C18 column (250 × 20 mm, 7 μm; Eka Chemicals, Bohus, Sweden) and a UV detector set at 250 nm. The analytical HPLC was performed on a Phenomenex (Torrance, CA, USA) RP–18 column (Jupiter 300, 150 × 2 mm, 3 μm) under same chromatographic conditions. The analytes were purified/separated, using gradient method. The mobile phase compositions were 0.1% HCOOH in Milli-Q water (Phase A) and 0.1% HCOOH in a mixture 50:50 (*v/v*) of CH₃OH and CH₃CN at volumetric ratio 1:1 (Phase B).

2.2. Materials

The benzene-1,3,5–tricarboxylic acid and benzene-1,2,4–tricarboxylic acid are Sigma–Aldrich Inc. products. They are used for the synthesis after chromatographic purification. The salts AgNO₃ and ZnCl₂ anhydrous are Sigma–Aldrich/Merck products as well. The catalysing processes are obtained according [1c] by mixing of 0.150–0.178 g of the acids preliminary dissolved in 25 mL solvent mixture CH₃OH:CH₃CN 1:1 with the TM salts. The molar ratio metal–to–ligand is 1:1 up to 4.7 in same solvent mixture. To these solutions are added drop wisely KOH (or NaOH) (1 M solution in CH₃OH or H₂O). The pH is controlled by the recalculation to the solvent mixtures. The deuterium –COOD labelled compounds were obtained using 37% solution of DCl in D₂O (99.5% Sigma–Aldrich/Merck), 30% solution of NaOD in D₂O (99% Sigma–Aldrich/Merck) and D₂O solvent (99% Sigma–Aldrich/Merck). The quaterphenyls [1,1'; 4',4'' 1'',1''']Quaterphenyl-2,4,5,2',3',5',2'',5'',3''',5'''-undecacarboxylic acid (**1**), 3'',4''',5'''-tris-dihydroxymethyl-[1,1'; 4',4'; 1'',1''']quaterphenyl-2,3,5,2',5',2'',5'''-octacarboxylic acid (**2**), 4,3',5'-tris-dihydroxymethyl-[1,1'; 4',1'; 2'',1''']quaterphenyl-2,5,2',4'',5'',2''',3''',5'''-octacarboxylic acid (**3**) and 9,9'-dioxo-9H,9'H-[2,2']bifluorenyl-1,3,5,7,4',6',8'-heptacarboxylic acid (**4**) were determined. The formation of oligomeric quaterphenyls is evidenced experimentally by independent UV–VIS analyses (See Fig. S7) exhibiting a band at λ_{max} ~220 nm, which is typical to substituted phenyl rather than to conjugated polycyclic

hydrocarbons. The low concentrations of the studied products of interaction prevent employment of IR and NMR methods for structural characterization. Moreover as it was shown in the section 'abstract' to this study there is needed further optimization of the reactions in order to achieve wanted to the field of organic synthesis yields.

2.3. Chemometrics

The chemometric analysis was performed by R4Cal Open Office STATISTICS for Windows 7 program package (<http://de.openoffice.org/>) [6(j,k)].

3. Theory/calculations

Quantum chemical computations were carried out by GAUSSIAN 98, 09; Dalton2011, Gamess–US, and GausView03 program packages [7a–d]. Thermodynamics were also calculated by Moltran v 2.5 program [7e]. The geometries of protonated peptides and their MS fragment ions were optimized employing *ab initio* and DFT methods. Becke 3–parameter exchange with Lee–Yang–Parr correlation (B3LYP), Becke 3–parameter46 Perdew–Wang–1991 (B3PW91), ω B97XD, QCISD and long–range corrected (LC)– ω PBE (Perdew–Burke–Ernzerhof) as well as Truhlar's M06, M06–2X and M06–HF functionals were used [7]. MP2 and 4 order ones were also utilized. The Truhlar's hindered rotor approximation method was employed [7i]. The Bernys' algorithm was utilized for ground state computations. The stationary points on potential energy hyper surfaces were obtained by standard analytical harmonic and anharmonic vibrational analysis. The absence of imaginary frequencies and negative eigenvalues of second-derivative matrix were used as criteria for energy minima. Rotational constants were extracted by optimized geometries. When used in internal energy determinations via Rice–Ramsperger–Kassel–Marcus (RRKM) calculations, frequencies were scaled by 0.99. Dunning's basis sets (cc-pVnZ) ($n = D, T$ and 5), 6–31++G(2d,2p), and quasi-relativistic effective core pseudo potentials from Stuttgart–Dresden–Bonn (SDD, SDDAll, <http://www.cup.uni-muenchen.de/oc/zipse/los-alamos-national-laboratory-lanl-ecps.html>) were employed [7]. Excitation energies were obtained by CIS, CIS(D) or time–dependent (TD)–DFT methods, using GDIIS optimization. The energy improvement was achieved by W1BD method [8], because of available references have shown that for organic compounds W1 provides more accurate data about ionization potentials (IPs), than Gn ($n = 2, 3$ and 4) and CBS methods [8a]. On the contrary W2 weak improves the electron affinities (EAs) values and insignificant perturbations of IPs values causing for an absolute mean error 0.23 kcal mol^{−1} [8b]. The Gn ($n = 4$) methodology reveals increasing in accuracy [8c] comparable with W1BD method. For adiabatic ionization potentials (AIP) it is $\in 0.00$ –0.04 eV [8c]. The deviation between theory and experiment studying EAs is $\in (-0.001)$ –(0.051) (W1), $\in (-0.001)$ –(0.039) (W2), $\in (-0.010)$ –(0.110) (Gn ($n = 2$)), $\in (-0.006)$ –(0.175) (Gn ($n = 3$)) and $\in (-0.009)$ –(0.121) eV (CBS–QB3) [8a]. We utilize, herein, W1BD and complete basis set methods (CBS) predicting AIPs. The mean absolute error is 53 kcal mol^{−1}, while the accuracy towards bond dissociation energy (BDE) – 1 kcal mol^{−1} (0.04 eV) [8q]. There was performed a correlative analyses between Koopmans', extended Koopmans' theorems; outer valence Green's functions (OVGF), equation-of-motion (EOM)–CC and P3 methods. The accuracy of IPs and EAs was compared with *ab initio* and DFT approaches, balancing between accuracy and computational cost. Non–diagonal GW approaches like NR2 and 3 + ones have been used [8e–h] as well, but they reveal an increasing in the computational cost. Moreover, the accuracy of OVGF is approximatively identical to

this one of NR2 studying IPs (<0.2 eV). Or the OVGF shows a high accuracy. The analysis includes coupled cluster doubles and perturbative triples (CCSD(T)) level of theories and basis sets aug-cc-pVnZ ($n = D, T, Q$ or 5). An increasing in accuracy of predicted geometry parameters of 0.002 Å was obtained by aug-cc-pV5Z basis set [8i,p]. On the other hand the DFT methods have shown higher accuracy in predicting energetics, including IPs, for instance, these are long–range corrected hybrid functionals (LC– ω PBE [8g]); hybrid functionals like B3LYP (exact exchange 20%), BHandHLYP (50%), M06–2X (54%) and M06–HX (100%) [8j]. The M06 data have shown that DFT approaches underestimate IPs [8j,k,m,o]. But a correlation analysis between IPs obtained at B3LYP/6–31G** and the experiment shows reliability 98.4% ($r^2 = 0.9843$) [8n]. The deviation from IPs to zero point energy (ZPE) is accounted for 0.3 eV [7e–h]. The computations in solution were carried out employing explicit super molecule and “mixed” approach of micro hydration and PCMs (CPCM and IPCM, IEPCM) [7]. The successful application of PCMs for accurate prediction of IPs by DFT approaches has been discussed [8j,l]. The largest species were treated by ONIOM method. Different fragments of a super molecule were computed using different accuracy, basis sets or methods. Molecular mechanics (MM)/molecular dynamics (MD) computations of interacting molecules were also used. The used MM methods were DREIDING and UFF force field ones [9]. The MD computations were obtained by classical approaches, *ab initio* Born–Oppenheimer molecular dynamics (BOMD) and atom centred density matrix propagation (ADMP) approach. The BOMD was carried out at M06 and M062x functional as well as SDD and cc-pvDZ basis sets without to consider periodic boundary condition. The electrostatic potentials (ESPs) and natural bond orbital (NBO) charges were employed in predicting the protonation ability of the molecules/fragment ions [9]. The ESPs computations were carried out using Chirlian's–Francel's and Breneman's grids.

4. Results and discussion

4.1. Chromatographic data

The chromatographic analyses of the interacting systems 135BA/Ag^I–salt and 124BA/Ag^I–salt are depicted in Fig. 1. As it has been reported [1c] the major products of interactions of poly-functional benzoic acids are 6H-benzo-[c]-chromene[6]-one, chromeno[5,4,3-cde]chromene-5,10-dione, and 4,10-dioxapyrene-5,9-dione. In the cases of 135BA and 124BA they correspond to the chromatographic peaks at retention times (RTs) 2.73 and 6.08 min. As these analyses show within the RTs $\in 4$ –14 min there are found a series of peaks of minor products of interaction (<6%) which are object of a comprehensive mass spectrometric and quantum chemical treatment in this paper. In order to determine the number of the chromatographic components we carry out chemometric processing of the chromatographic patterns (Fig. 1 and S1, Table S2). There are found peaks at RT = 4.64, 4.98, 5.06, 10.05, 10.76, 11.23 and 12.26 min in 124BA/Ag^I–salt system. While the corresponding data for system 124BA/Ag^I–salt are 7.37, 8.09 and 8.45 min, respectively. In particular, the attention is focused on closely disposed peaks at RT = 7.43 (124BA/Ag^I–salt) and 7.37 min (124BA/Ag^I–salt) in both systems in order to gain the information is Ag^I–catalysing process with participation of different positional isomers can be regarded, in parallel, as a regioselective one.

4.2. Mass spectrometric data

4.2.1. Structural analysis

Fig. 2 reveals the ESI–MS spectrum of 124BA/Ag^I–salt system at RT = 12.26 min exhibiting a MS peak at m/z 789 of monocation of a

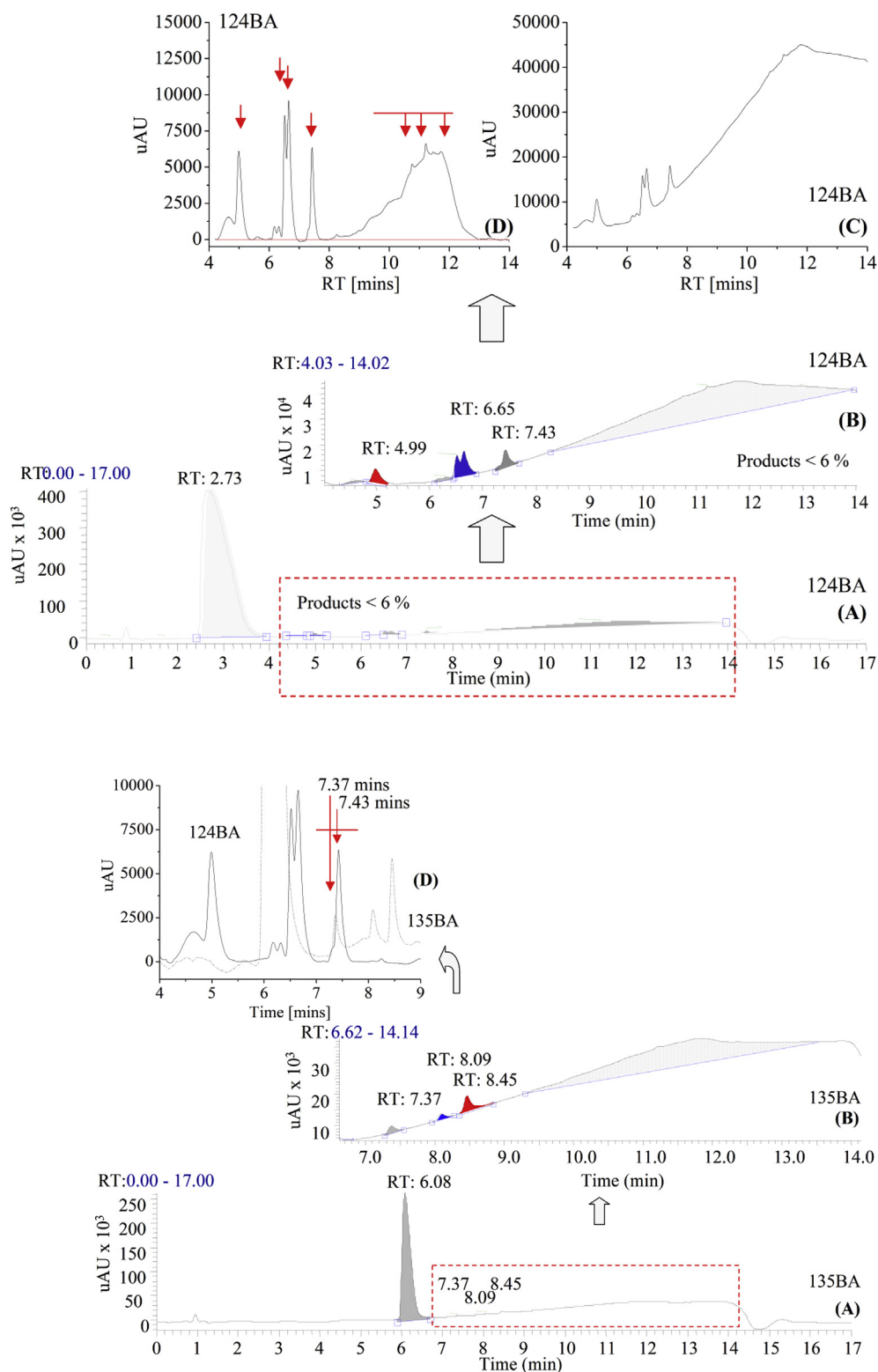


Fig. 1. Chromatograms within the framework of retention times $RT \in 0-17$ min of the products of interaction of benzene-1,2,4-tricarboxylic acid (124BA) and benzene-1,3,5-tricarboxylic acid (135BA) in presence of Ag^I -ions (A); Chromatograms within $RT \in 4-14$ min (B); Non mathematically processed chromatogram of 124BA (C) and chromatograms after an application of baseline correction approach (D).

functionalized quaterphenyl $\{[C_{35}H_{17}O_{22}]^+\}$, i.e. [1,1'; 4',4'; 1'',1'''] Quaterphenyl-2,4,5,2',3',5',2'',5'',2''',3''',5'''-undecacarboxylic acid (**1**). The subsequent decarboxylation reactions have been found as typical studying MS spectra of monomeric and dimeric benzoic

acids as well as diphenyl [1c,10]. Our data allow us to conclude that the same phenomenon is typical for the quaterphenyls as well. Furthermore, there have been produced publications indicating that the discussed process is accompanied with possible

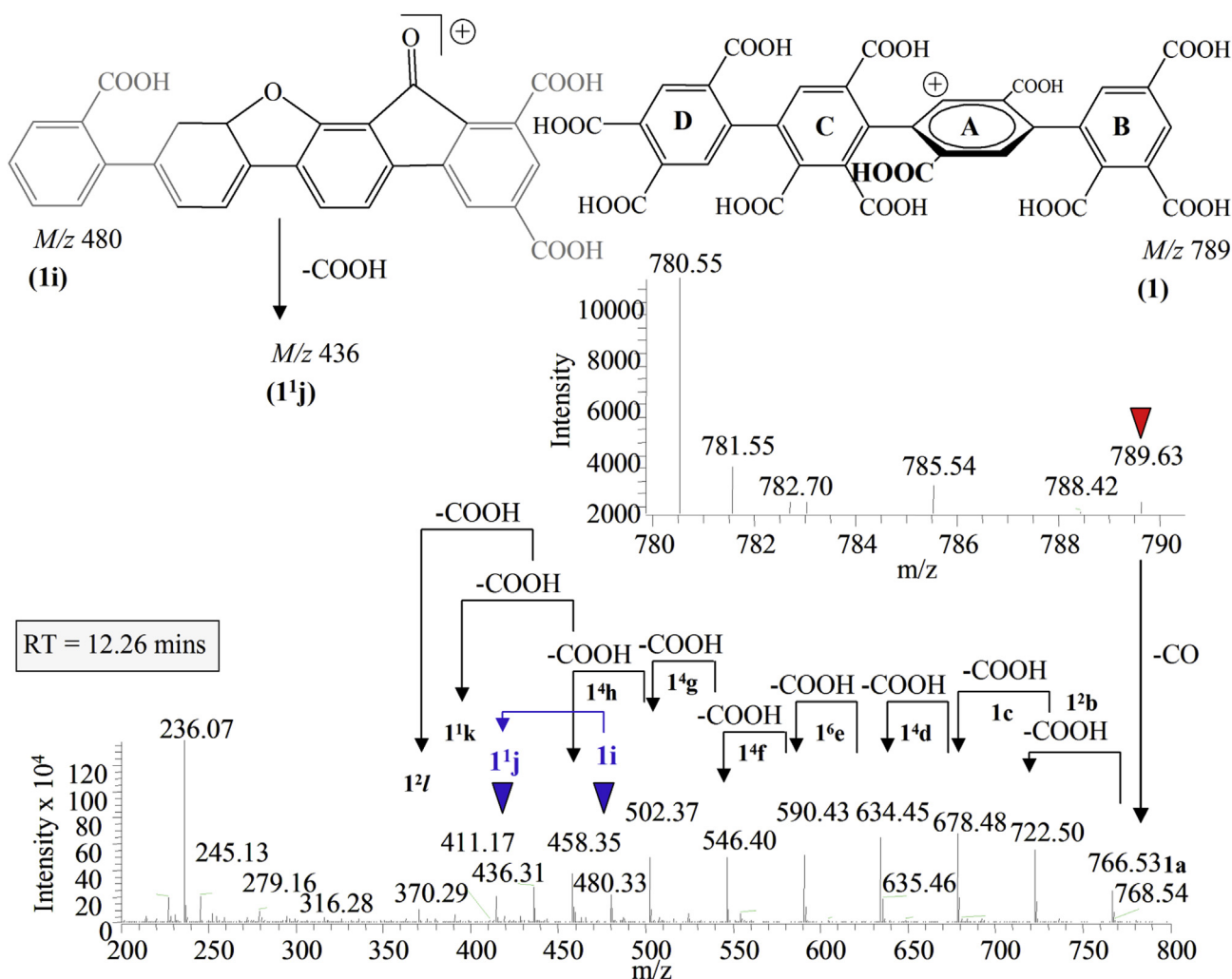


Fig. 2. Experimental ESI–MS and CID–MS spectra of (1)–(3) at RT = 12.26, 12.01 and 11.85 and of isotope labelled product of organozinc(II) catalysed reaction of 135BA at m/z 783; Chemical diagram of the fragment ions and the decarboxylation MS ions.

intramolecular rearrangement and migration of the carboxylic–OH group and in parallel a loss of CO–functional fragment [10]. The process cause for formation of substituted benzo furans [10]. In addition there can be observed a direct decarboxylation (or loss of –COOH group); loss of a single proton or –OH group. It is important to note, however, that on the contrary the MS reactions of monomeric benzoic acid derivatives, where the loss of –OH group is accompanied with formation of phenylmethyldiyl-oxonium or cyclohexa-2,4-dienylidenemethylene-oxonium cations, thus further participating into reactions of loss of CO group [11], the oligophenyls tolerate C–C bond formation reactions instead of the latter process. So far, we have discussed at least three important aspects of the gas – phase MS reactions of benzoic acids and their derivatives which contribute to the MS profile of the highly functionalized oligomeric quaterphenyls. Experimental MS studies support the formation of benzo[c]chromen-6-ones as well [1c,10]. In the case of the oligomer (1) (RT = 12.26 min) we can argue that the observable MS isotope shape with highest intensive peak at m/z 766 from the experimental ESI–MS spectra support the model of above–mentioned intramolecular rearrangement producing cationic 3-(2,4,5-Tricarboxy-phenyl)-7-(2,3,5-tricarboxy-phenyl)-4,4a,5a,8,9,9a-hexahydro-ibenzofuran-1,2,6,9-tetracarboxylic acid (1a) (m/z 766), which further decarboxylation causes for isolation of

cationic 3-(2,4,5-Tricarboxy-phenyl)-7-(2,3,5-tricarboxy-phenyl)-4,4a,5a,8,9,9a-hexahydro-dibenzofuran-1,2,6-tricarboxylic acid (1b) (m/z 722), etc. ions (1ⁱc)–(1ⁱl) (i = 1–8) shown in Fig. 2 and Scheme 1. It is important to mention that as a result of the poly-functionalized character of the studied quaterphenyls there are described from a thermodynamic point of view all possible mechanisms of decarboxylation in order to determine the correlation between the molecular structure and the thermodynamic stability of the fragment species (See the sub–section 3.3, Fig. S2). The latter knowledge plays a crucial importance deducing the fragment mechanisms for the complex analyte molecules from experimental mass spectrometric data. However, as it was discussed above the detail understanding of the role of the molecular functionalization in the regioselective reaction mechanisms contributes significantly to the molecular design of organometallic catalysts for goal oriented organic synthesis of efficient, and in parallel, high regio (enantio) selective approaches. Thus the fragment pathway of (1) and the corresponding substituted substituted 9-phenyl-10a-dihydro-11-oxa-indeno[2,1-a]fluoren-12-one (1i) involve (1) → 1a → 1²b → 1c → 1⁴d → 1⁶e → 1⁴f → 1⁴g → 1⁴h → 1¹k → 1²l and 1i → 1²j, respectively. As the thermodynamic data in the next sub–section show the decarboxylation reactions within the framework of the substituted D–aromatic ring are, in general, favourable from

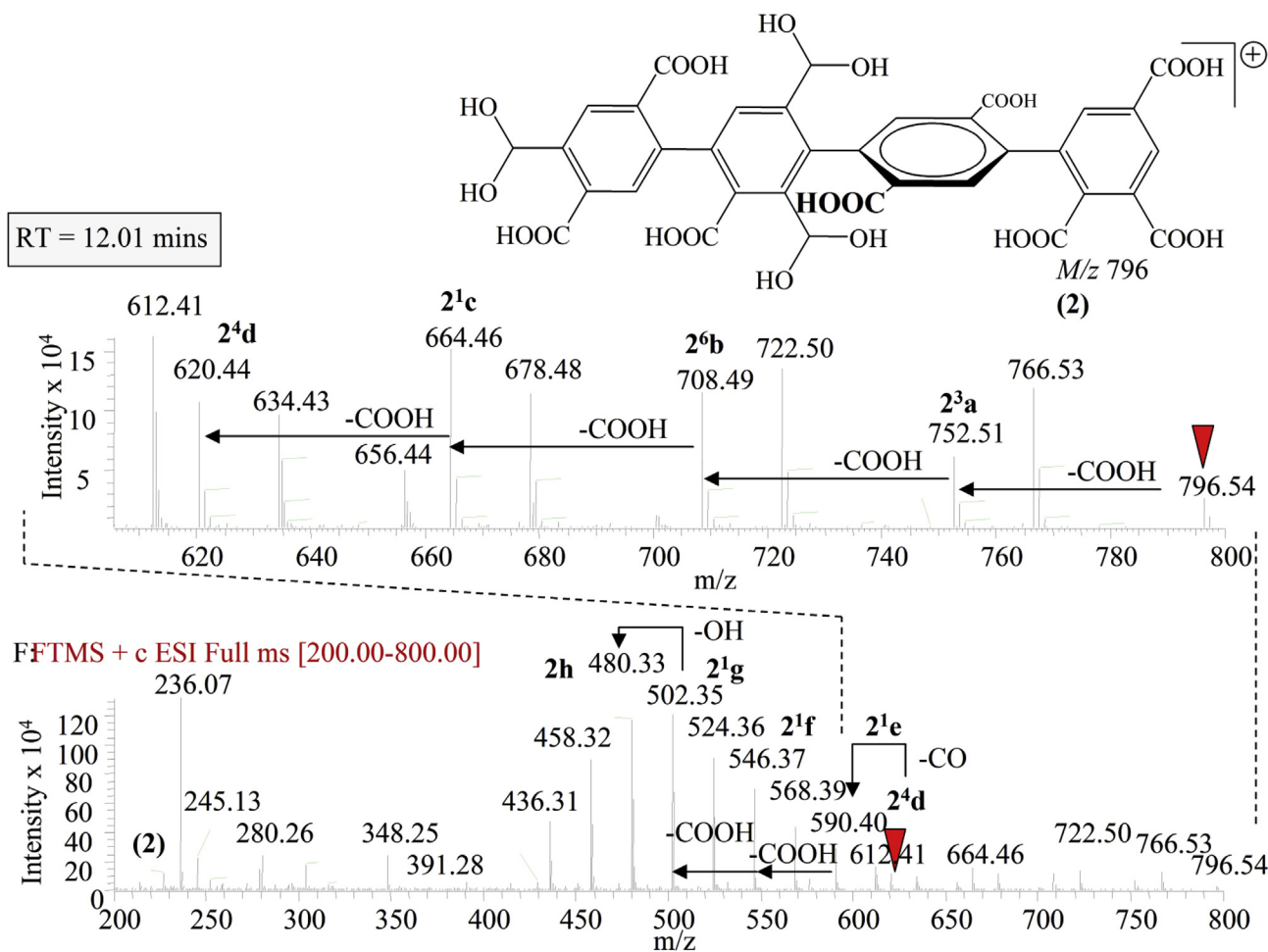


Fig. 2. (continued).

the energetics point of view. Fig. 2 reveals as well as ESI–MS fragment patterns of oligomer products at RT = 12.01 (2) and 11.85 (3) mins. As can be seen the latter MS fragment reactions produce different fragment within values $m/z \in 400$ –800 of (3) comparing with MS shapes of 3'',4'',5''-Dihydroxymethyl-[1,1'; 4',4'; 1'',1'''] quaterphenyl-2,3,5,2',5',2'',2''',5'''-octacarboxylic acid (2) which are affected from the different type substitution from the major fragment pathways of the analogous derivative (1) discussed above. The differences in the reaction kinetics and thermodynamics between (1) and (2) are illustrated in the next sub-section 3.2.2 dealing with the quantification of MS intensities of fragment ions. Here, we present only the different MS ions, mainly found within $\in 600$ –800 (Fig. 2) in the light of fragment processes originating due to the effect of the different substituents at 3'',4'',5''-positions in (2). The decarboxylation reactions of (2) cause for the formation of a set of ions via the pathway (2) $\rightarrow$ $2^3a \rightarrow 2^6b \rightarrow 2^1c \rightarrow 2^4d$. The products of interaction (2) and (3) are obtained due to 2- and 4-decarboxylation reactions accompanied with C–C bond formation (below, Fig. S3). As positional isomers there is expected as a significant difference in the corresponding MS spectra as evidenced in Fig. 2. In order to obtain the proton position of the cation m/z 797 we perform NBO analysis (Fig. 2) on which base there is obtained the shown fragment pathway. The $q_O(\text{NBO})$ value of -0.27642 is obtained for the O-centre of the depicted in the latter figure chemical diagram of the substituted quaterphenyl (3). Here, we should stress that the formation of germinal diols from carboxylic

acids is in generally known [11]. The explanation is concerned in electronic rearrangement affecting mainly ortho-positions in the phenyl ring. The comprehensive analysis of deuterium exchange reactions in labelled benzoic acids has been shown that this process appears leading one causing for a deuterium exchange reactions of the labelled COOD with the proton at o-position [11].

The decarboxylation of protonated $-\text{COOH}_2^+$ group leads to the cationic ($3a$) fragment (m/z 752). While the further loss of $-\text{COOH}$ functional group is associated with formation of (3^3b) cation (m/z 708). Fig. S4 and Schemes S1 and S2 provide an excellent deal of evidence to correlation between thermodynamic energetics of (i^jb) cations (i = compounds (2) or (3); j = 1–7 decarboxylated forms) obtained by MD computations and the experimental MS intensities of the MS peaks at m/z 708 of both type substituted oligomers as far as the intensities in case of (3) are significantly higher comparing with those in compound (2). However, there are not many reports which examine correlation between electronic structure of MS cations and intensity of the corresponding MS peaks. However, our results show that the decarboxylation reactions in A-phenyl ring in (3^jb) series of cations are more favourable from a thermodynamic point of view comparing with the process of loss of $-\text{COOH}$ in phenyl rings C- or D-one respectively. On the contrary the (2^jb) series of cations reveal an opposite behaviour of relationship between total energy (E^{TOT}) as a function of the ring type decarboxylation reactions describing the loss of $-\text{COOH}$ groups in phenyl A–D rings. As the major intramolecular factors causing the greatest

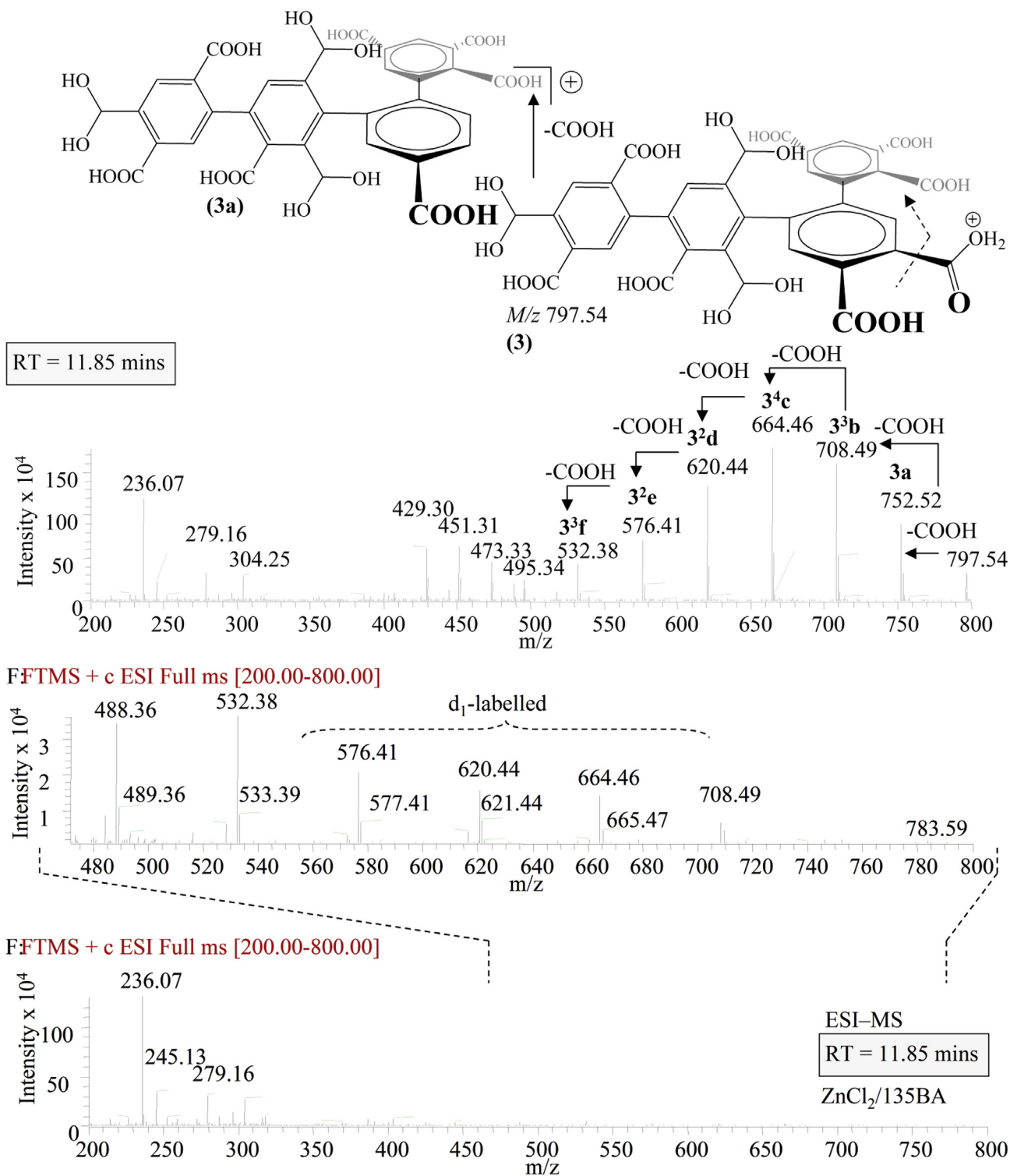
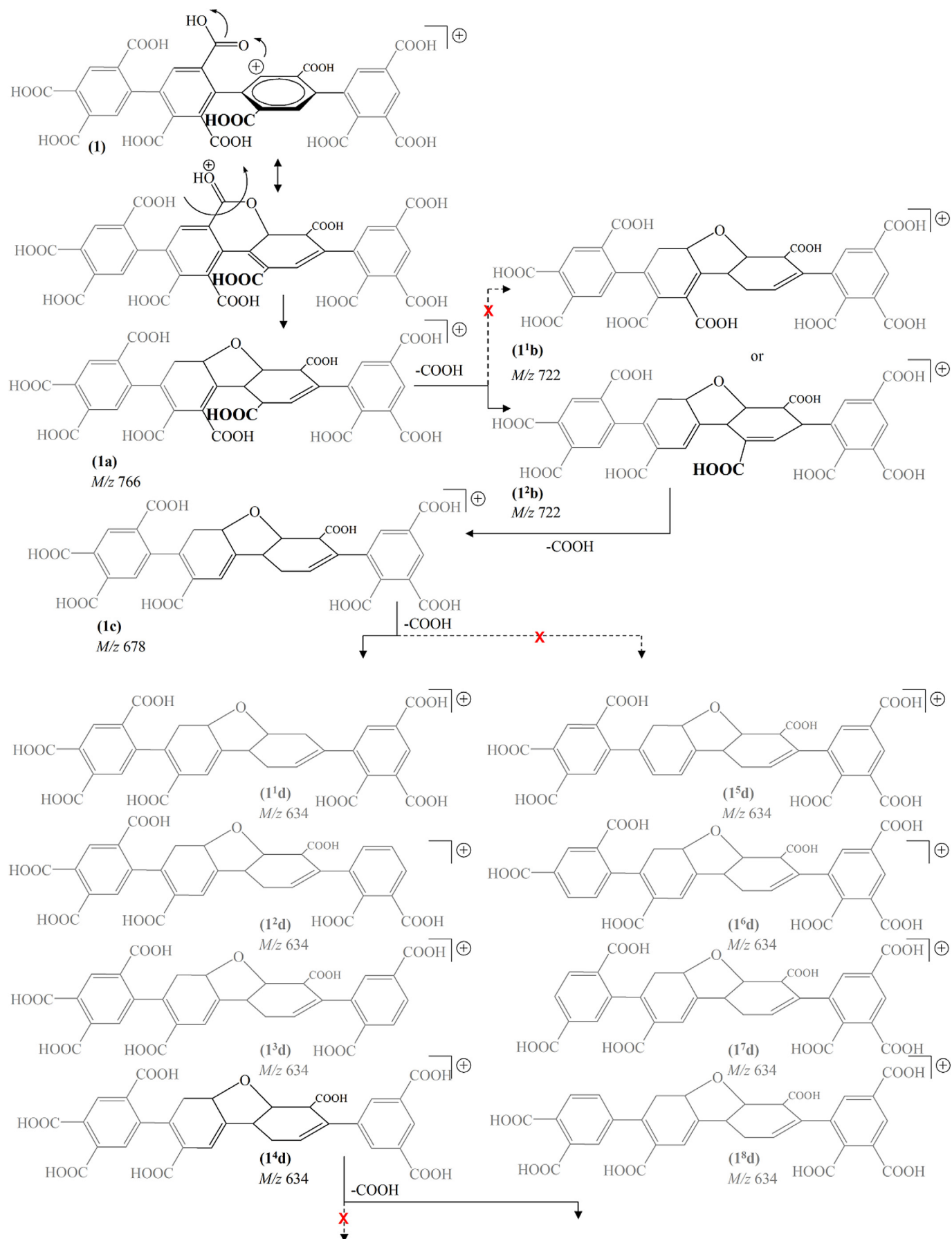


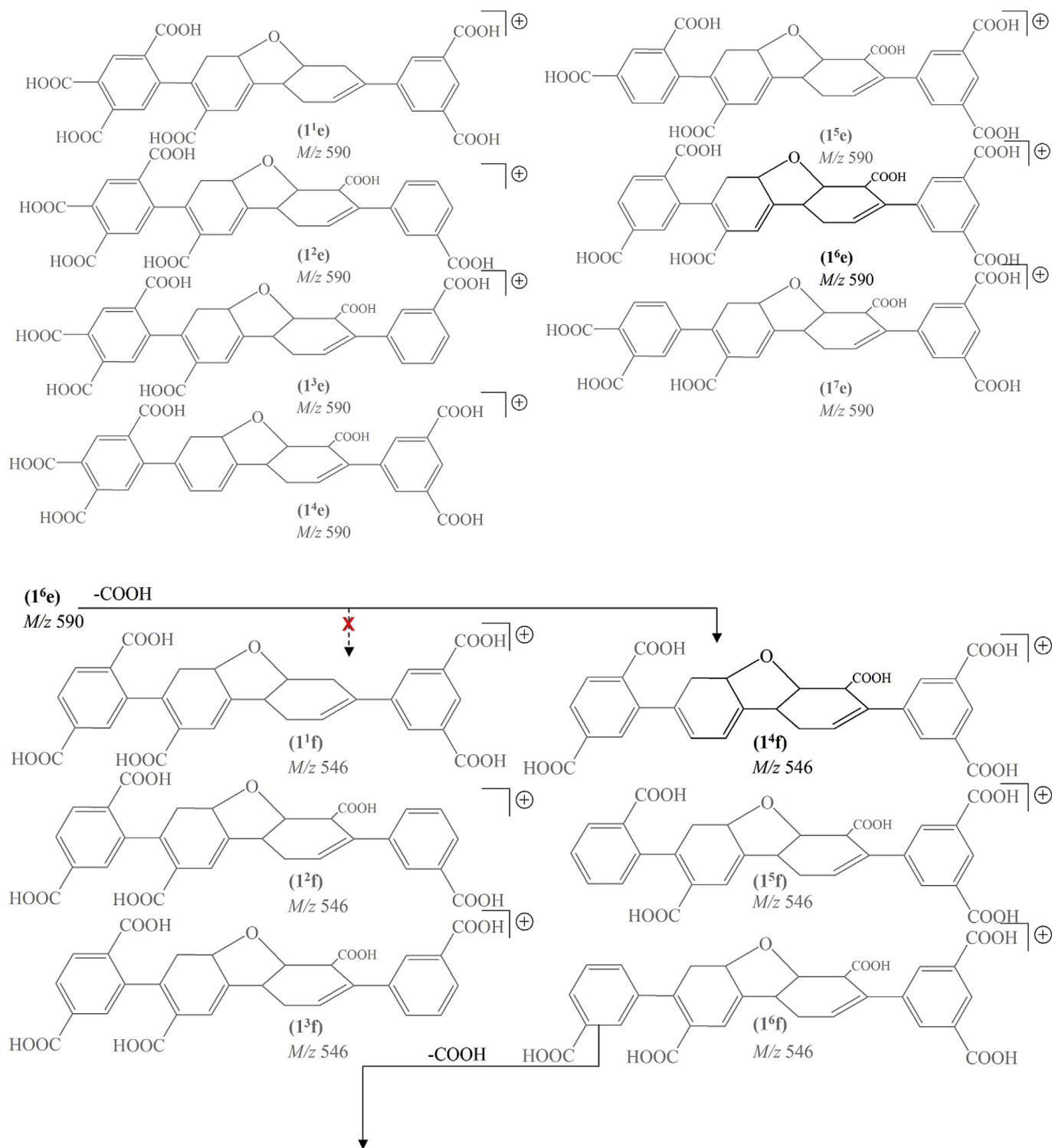
Fig. 2. (continued).

thermodynamic stability of the fragment ion (**3^{3b}**) are the short contacts $r(\text{O}\cdots\text{O})$ (3.499 and 3.516 Å) and moderate $\text{OH}\cdots\pi$ interaction (2.761 Å) involving the germinal diol of the D-ring and π -aromatic system of A-ring (Scheme S3). The fragment pathway (**3a**) $\rightarrow$ **3^{3b}** $\rightarrow$ **3^{4c}** $\rightarrow$ **3^{2d}** $\rightarrow$ **3^{2e}** $\rightarrow$ **3^{3f}** is realized. The CID–MS fragmentation patterns of (**1**) are governed by cleavage of $-\text{OH}$ groups as shown in Fig. 3 studying fragment reactions of cations **1^{4g}** (m/z 502) and **1^{6e}** (m/z 590), respectively. Fig. 3 and S5 depict in parallel

to experimental CID–MS spectra the chemical diagrams of the molecular ions, their energies and $q_0\text{NBO}$ charges on which base we have concluded about the fragment pathways **1^{4g}** $\rightarrow$ **1^{4g3a}** and **1^{4h}** $\rightarrow$ **1^{4h1c}**, respectively. A fragment reaction **3^{2e}** $\rightarrow$ **3^{2e1d}** is obtained, as well. As is well known the isotope labelling techniques appears the most useful MS tool for elucidation of reaction mechanisms especially studying processes of carboxylic acid derivatives tolerating, in general, proton rearrangement reactions as shown



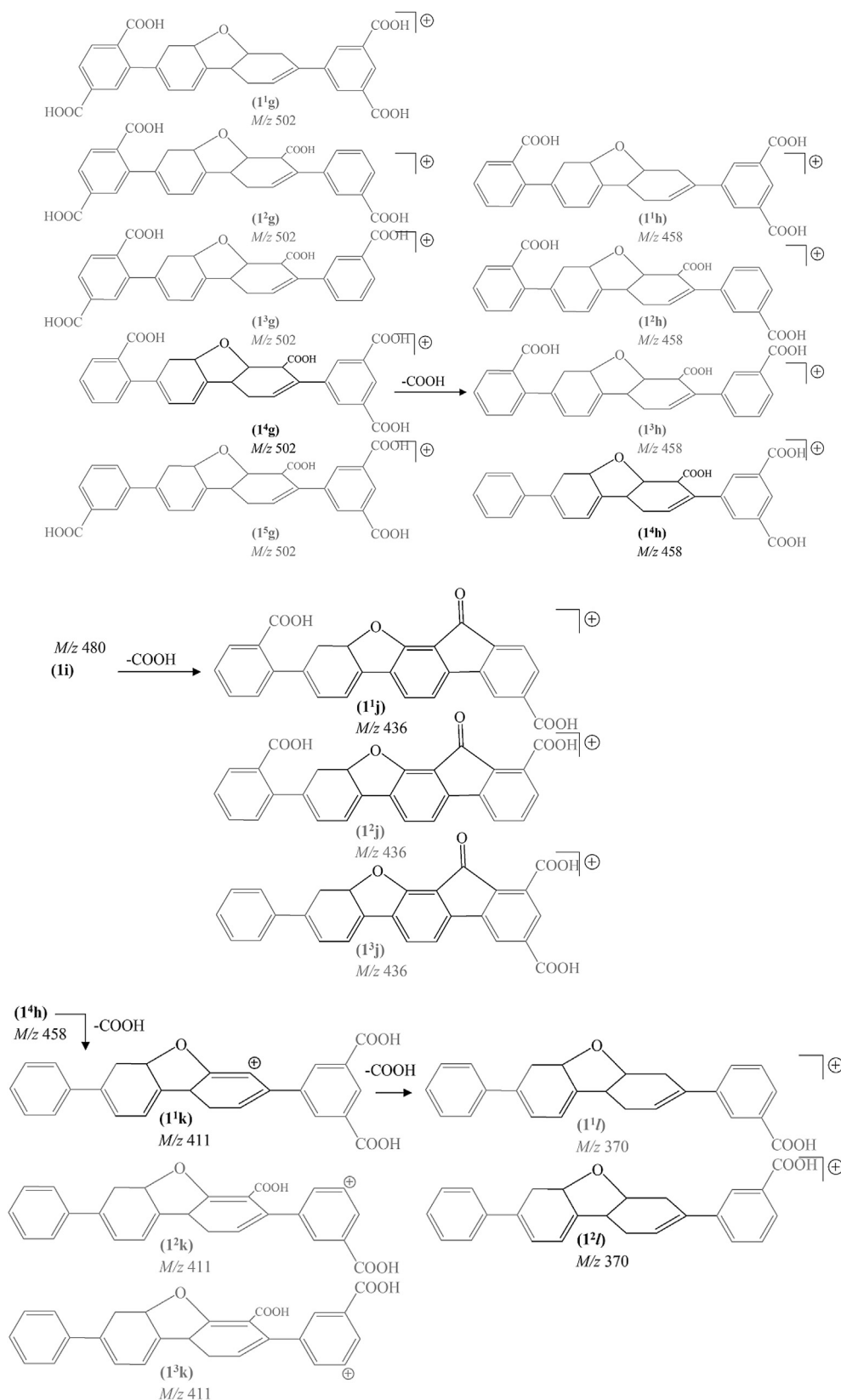
Scheme 1. Chemical diagrams of MS fragment ions of (1) due to decarboxylation reactions and the corresponding substituted 9-phenyl-10,10a-dihydro-11-oxa-indeno[2,1-a]fluoren-12-one (1i); Mechanistic aspects of CO-loss in the case of (1) (Continue as Schemes S1–S5).



Scheme 1. (continued).

above [11]. It needs perhaps to be pointed out clear that the interpretation of the isotope labelled compounds is difficult due to possible deuterium scrambling, however. On the other hand, it is worthy nothing that there is a great variety of already studied carboxylic acid derivatives so that certain aspects of the typical MS reactions affecting the deuterium labelled carboxylic acids are already well understood [11]. More detail can be found in Ref. [11d] on this base we perform analyses of highly functionalized

deuterium labelled carboxylic acid oligomers. The CID–MS spectrum of isotope labelled (3) is depicted in Fig. 4. The MS peak at m/z 451 belongs to (3g) cation exhibiting a -COOH_2^+ charged fragment. A more detail information about the proton accepting ability of (3g) is obtained from the looking at $q_0(\text{NBO})$ values. The -OH group showing the lowest $q_0(\text{NBO}) = -0.726$ value. The fragment pathway (3g) $\rightarrow$ (3h^{1a}) + (d₁-3h^{2a}) occurs (m/z 386 and 387). As explained in the section 'Introduction' there are clear advantages of



Scheme 1. (continued).

computing the free Gibbs energies of the MS fragment ions in the way done in this paper, because of it has power in predicting very

sensitively the position of the isotope labelled atoms as well (see the next Section 3.3).

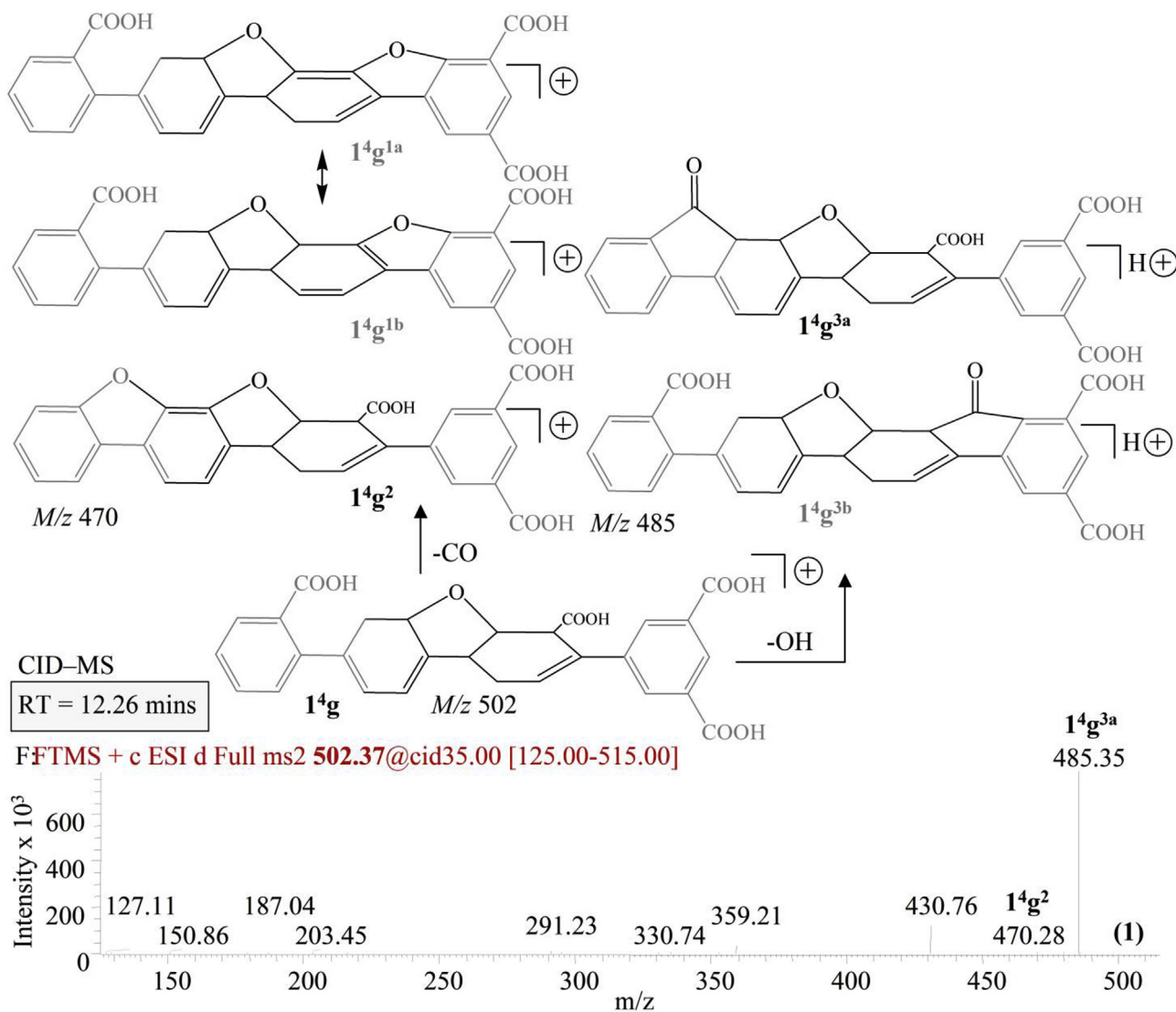


Fig. 3. CID-MS reactions of organosilver(I) catalysed products (1) and (3) (RT = 11.85 and 12.26 min) of ion at m/z 458, 576 and 590; Chemical diagram of the fragment ions; Theoretical NBO charges $q_0(\text{NBO})$ showing the probable proton accepting positions of the fragment cations.

In what follows in this sub-section, we will discuss on the great selectivity of the CID-MS method. In this respect, our study serves as a very illustrative example of the great ability to produce quantitative results depending on the analyte, in particular, talking about small changes of the molecular structures of the analytes. Accordingly in the experimental design it is taken into consideration the products of interaction of quaterphenyls of positional isomers of 1,2,4- and 1,3,5-benzene tricarboxylic acids (Figs. 5 and 6). The MS spectra at RT = 12.65 min are discussed. In the case of benzene 1,2,4-tricarboxylic acid derivatives the MS spectra belong to the hydrated derivative of (5), while in the case of benzene 1,3,5-tricarboxylic acid derivatives there is obtained 5,9-dioxo-2,6-bis-(2,4,6-tricarboxy-phenyl)-5H,9H-4,10-dioxo-pyrene-1,3,7-tricarboxylic acid (8). Both compounds (5) and (8) are obtained due to one and the same type organosilver(I) catalysed interaction of different substituted biphenyls. In this context there can be seen Scheme S4 and Fig. 7. The latter one reflects the MS spectra of a proposed organosilver(III) intermediate, which has been detected at RT = 11.45 min in system 124BA/Ag. We do not only account for the similarities and the differences of the fragment MS patterns in

qualitative terms (See Figs. 5 and 6), but also consider the quantitative relationships between the reaction kinetics and thermodynamics parameters in order to illustrate the great reproducibility of ESI-CID-MS experiment, as well (See the next Sub-section 3.2.2). Our 'Result and discussion' section first of all turns the MS fragment phenomena that do the ESI-MS experiment. Within small m/z values ($m/z \in 200-400$) both compounds (5) and (8) exhibit virtually similar MS profiles (Fig. 5), due to same major quaterphenyl molecular skeleton. On the contrary the MS profiles at $m/z > 400$ reveals the differences of the fragment pathways due to different type of positional substituted -COOH functional groups in the major [1,1'; 3',4'; 1'',1''']quaterphenyl skeleton. On the base on the respective CID-MS spectra at m/z 256 we found exact experimental evidences about the great selectivity of the method allowing to distinguish unambiguously the structurally positioned isomers (Fig. 6). A detail perspective on the great capability of CID-MS for structural analysis and quantification of analytes with very complex molecular scaffold can be obtained from the data in Fig. S6 looking at the fragment patterns of hydrated derivatives of (8), labelled as (8a), and (6). Like the MS profiles of (8) and (5)

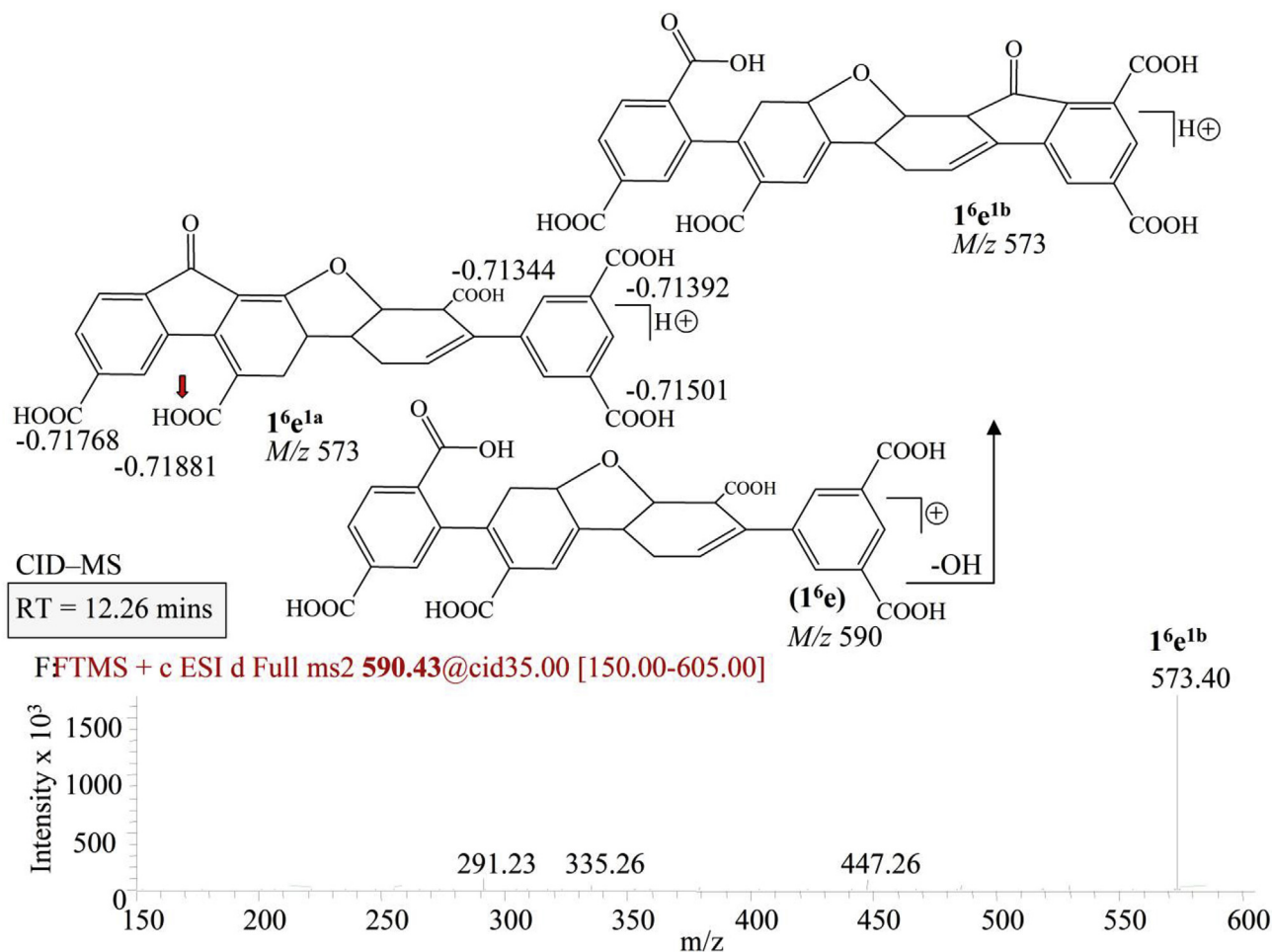


Fig. 3. (continued).

obtained under ESI(+) experiment, the corresponding data about (**8a**) and (**5**) show a virtual similarity of the MS patterns within $m/z \in 200\text{--}400$. However, unlike ESI data, CID–MS ones reveal a significantly informative picture. The advantages include qualitative and quantitative treatment allowing an unambiguous assignment of the fragment ions and obtaining of experimental kinetic and thermodynamic parameters (See the next sub–section and reference [120]). The CID–MS spectrum of dimeric substituted quaterphenyl 2-[4'-(4,5-dicarboxy-9-dihydroxymethyl-12-oxo-12H-11-oxa-indeno[2,1-*a*]fluorene-2-yl)-biphenyl-4-yl]-12-oxo-12H-11-oxa-indeno[2,1-*a*]fluorene-4,5,9-tricarboxylic acid (**7**) from (**2**) is shown in Fig. S6 along with the fragment scheme and the energies of the MS fragment ions as depicted in Scheme S1 and Fig. S3. The corresponding formation of trimeric substituted biphenyl from (**2**) is evidenced in the same Fig. S6. As it was studied in Ref. [1c] the Ag(I) and Zn(II) ions can catalysed one and the same type of fragment reactions of polysubstituted benzoic acids, thus yielding to corresponding biphenyls. Thus line of our experimental design illustrates, among others, that the statement above is valid to the reactions of quaterphenyls as well. Fig. 2 and S6 depict the MS spectrum of substituted quaterphenyl obtained due to organozinc(II) catalysing reaction of 135BA as well. The quantitative kinetics relationships from the corresponding CID–MS spectra are discussed in the next sub–section.

4.2.2. Quantitative analysis

In order to get a quantitative information about the level of “similarity” between the MS patterns of the substituted quaterphenyls obtained due to interaction of positional different isomers of benzene 1,3,5– and 1,2,4–tricarboxylic acid derivatives (**5**), (**6**), (**8**) and (**8a**) depicted in Figs. 5 and 6 we obtain experimental kinetic and thermodynamic parameters of the MS fragment reactions using the formalism described in Ref. [12a]. The relationship between MS intensity of peak of ionic species is given by equation (1) according [12]. The ratio of intensities of cation at time $\tau = t$ ($I_{M,t}$) towards intensity at $\tau = 0$ ($I_{M,0}$) – survival yield – represents function of dissociation rate constant $k(E)$ and the parameter τ , respectively. The latter quantity is time scale of experiment (in our case $\tau = 30 \text{ min} = 1800 \text{ s}$). The $k(E)$ is a function of internal energy of molecular species.

$$\frac{I_{M,t}}{I_{M,0}} = e^{-k(E) \cdot \tau} \quad (1)$$

According the equation model of Iribarne-Thomson [12], the relationship of the rate constant (k) is given by the Eyring equation (2).

$$k = \frac{k_B \cdot T}{h} \cdot e^{-\frac{\Delta G}{RT}} \quad (2)$$

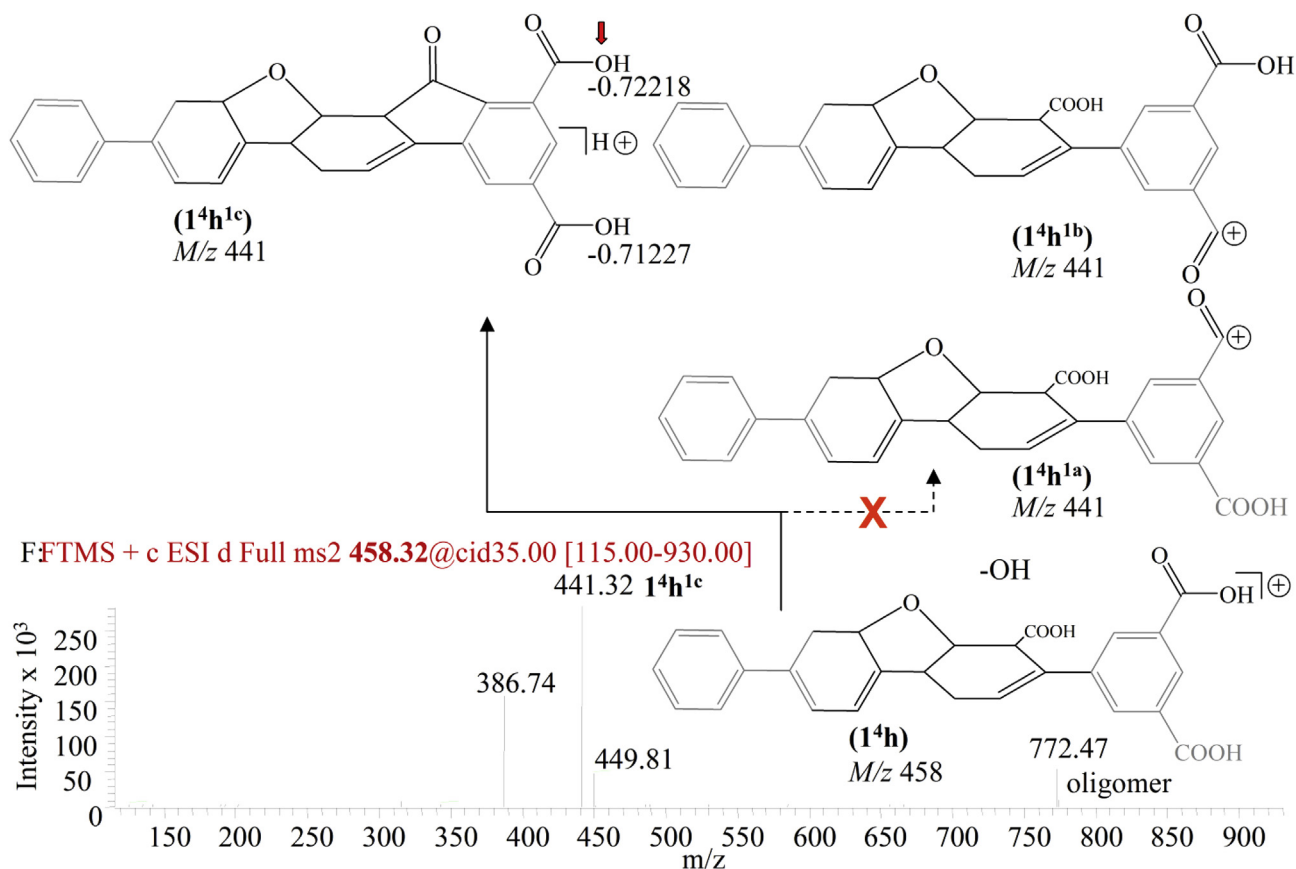
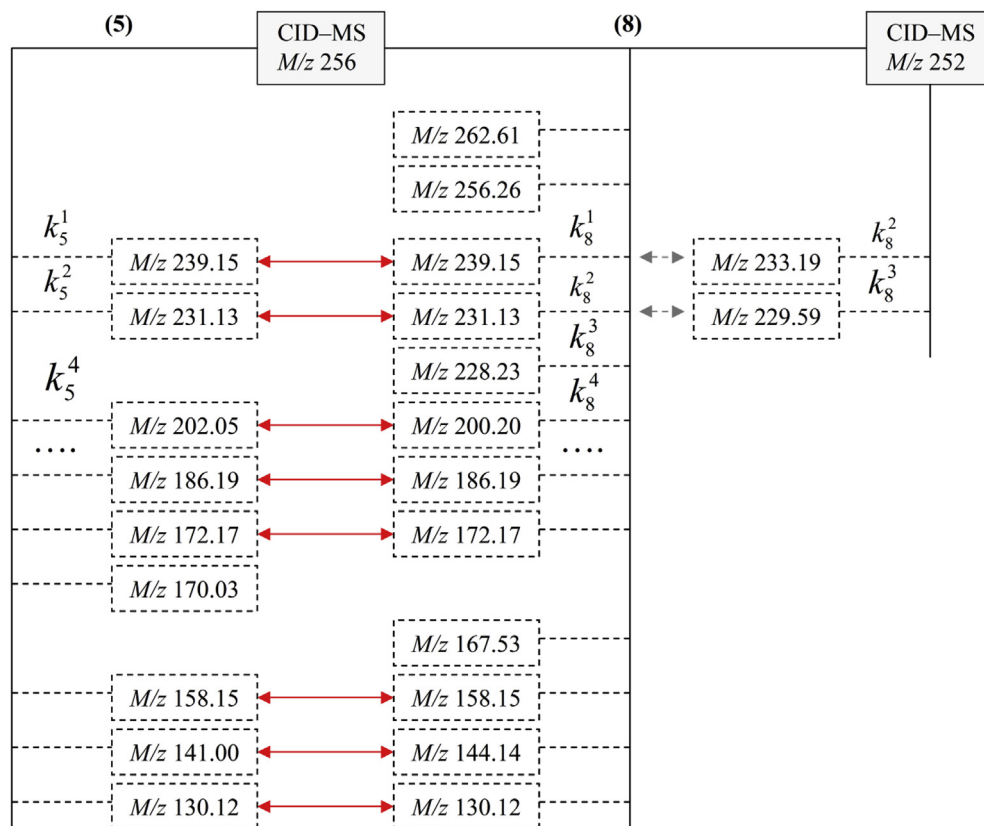


Fig. 3. (continued).

The collision energy is transformed into kinetic energy [12j]. The correlative analysis involves the following set of rate constants of

reactions leading to MS ions with virtually same m/z values ((5) and (8), Fig. 5):



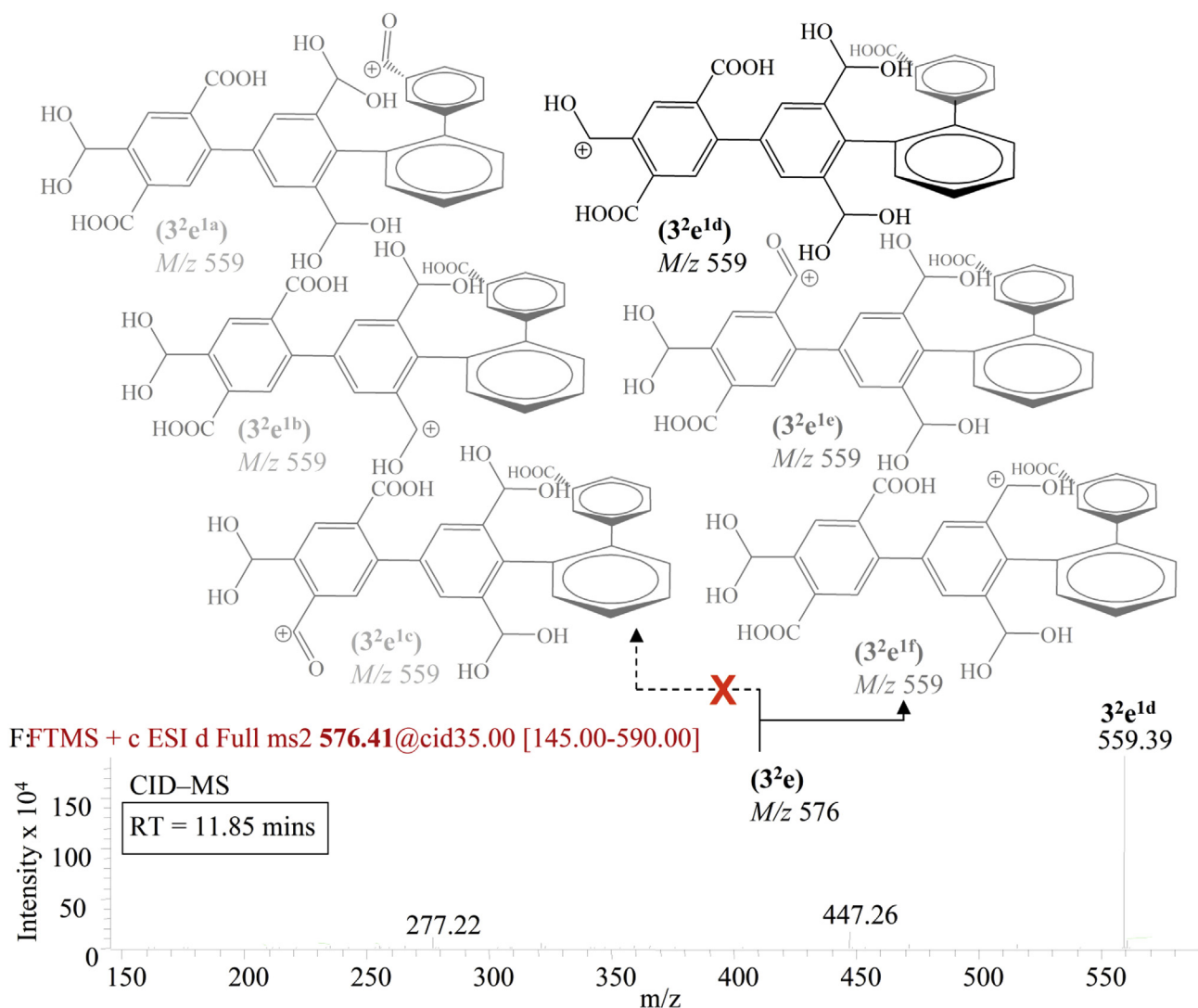


Fig. 3. (continued).

The k_{ij} rate constants of the corresponding to the i th fragment reaction under CID-MS conditions applied to the MS peak at m/z 256 of the j th compound for compounds (**5**) and (**8**) are: $k^1_5 = 0.00108287$, $k^2_5 = 0.0021980089$, $k^4_5 = 0.0034775947 \text{ s}^{-1}$; $k^1_8 = -2.45 \cdot 10^{-4}$, $k^2_8 = 3.355 \cdot 10^{-4}$, $k^3_8 = 0.0011015$ and $k^4_8 = 3.6917 \cdot 10^{-4} \text{ s}^{-1}$, respectively (Fig. 8). As might be expected the correlations between k^1_5 and k^1_8 rate constants are not significant, nevertheless that we treat MS fragment species with virtually similar m/z values, because of these latter are obtained as product ions within of different as electronic structures initial MS ions determined from the different disposition of the $-\text{COOH}$ substituents. The chemometric data can be found in the same Fig. 8. The level of confidence is 75.82% using a linear approximation. The corresponding kinetic parameters of the CID-MS reaction of (**8**) applied to MS peak at m/z 252 are: $k^3_8 = 7.366 \cdot 10^{-4}$ and $k^4_8 = 0.0012394 \text{ s}^{-1}$, respectively (Fig. 8). On the contrary the comparative analysis of the rate constants of CID-MS reactions

within the framework of the same compound (**8**) to MS peaks at m/z 252 and 256 leading to observation of MS fragment ions at m/z 228.23 and 231.13 as well as 233.19 and 229.59 shown above (See the CID-MS spectra in Fig. S6), where a weak perturbation of the electronic structures of the reaction ions occurs, results to a standard deviation value of k_{38} rate constants of both CID-MS reactions $\text{SD}(y_{\text{Er}}) = 9.75 \cdot 10^{-5}$ (Fig. 8, dotted circle). Or as might be expected the latter data correlate significantly. The examining of the reaction kinetics of $1^6\text{e} \rightarrow 1^6\text{e}^{1b}$ fragment process shows $k = -0.00360686 \text{ s}^{-1}$. Or the experimental ΔG value is $-20.85 \text{ kJ mol}^{-1}$ (-0.007943 a.u. , Fig. 3). The CID-MS spectra of the products of organosilver(I) and organozinc(II) catalysed reactions of 135BA at $\text{RT} = 11.85 \text{ min}$ (Figs. 2 and 6). In connection with the topic of the aforementioned quantitative statistically significant correlations of the reaction rates of fragment CID processes, let us then consider these relationships in the two independent Ag^{I} - and Zn^{II} - catalysed reactions of 135BA yielding to the chromatographic peak at $\text{RT} = 11.85 \text{ min}$. The experimental CID spectra

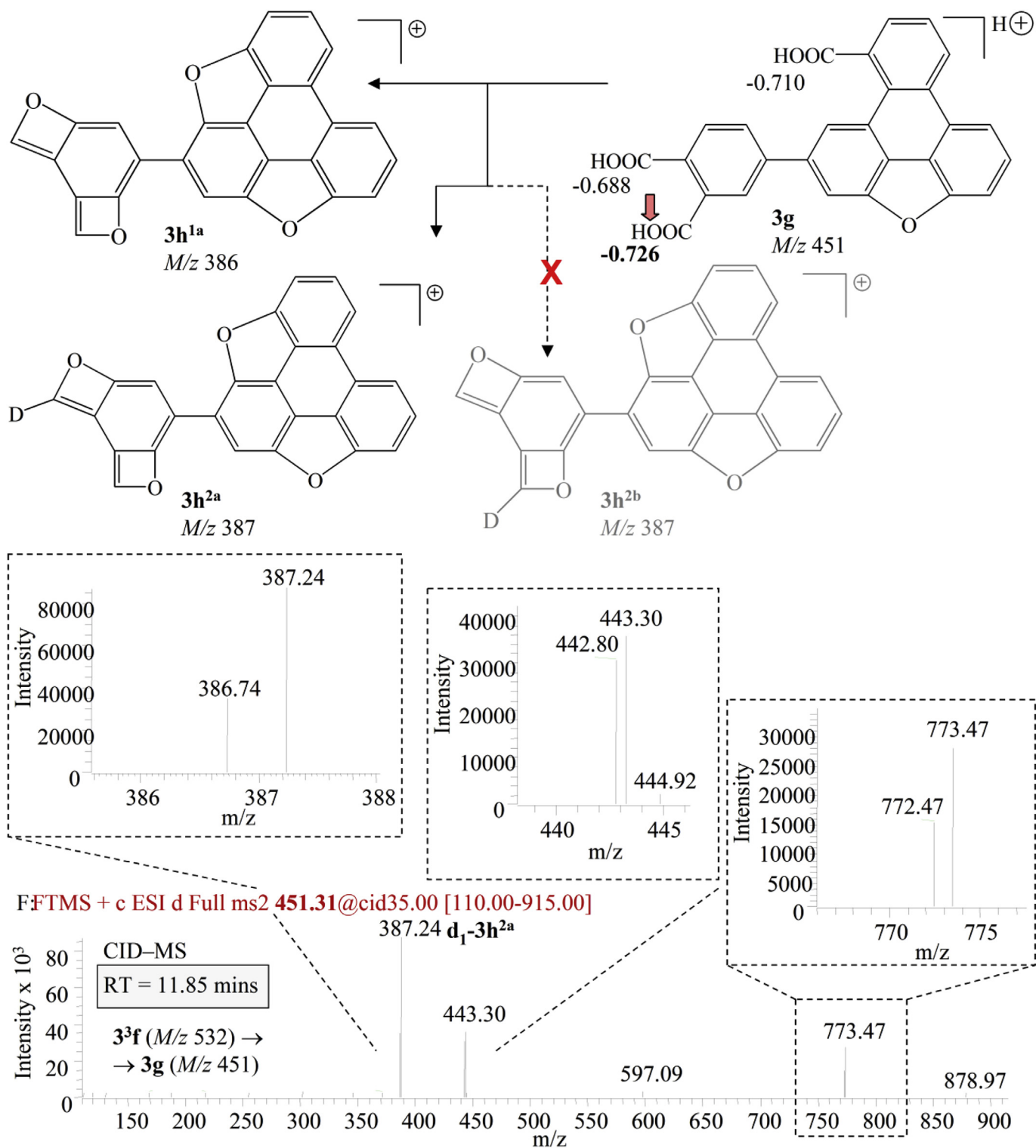


Fig. 4. Experimental ESI-MS spectrum of d_1 -(3); Chemical diagrams of the fragment MS ions.

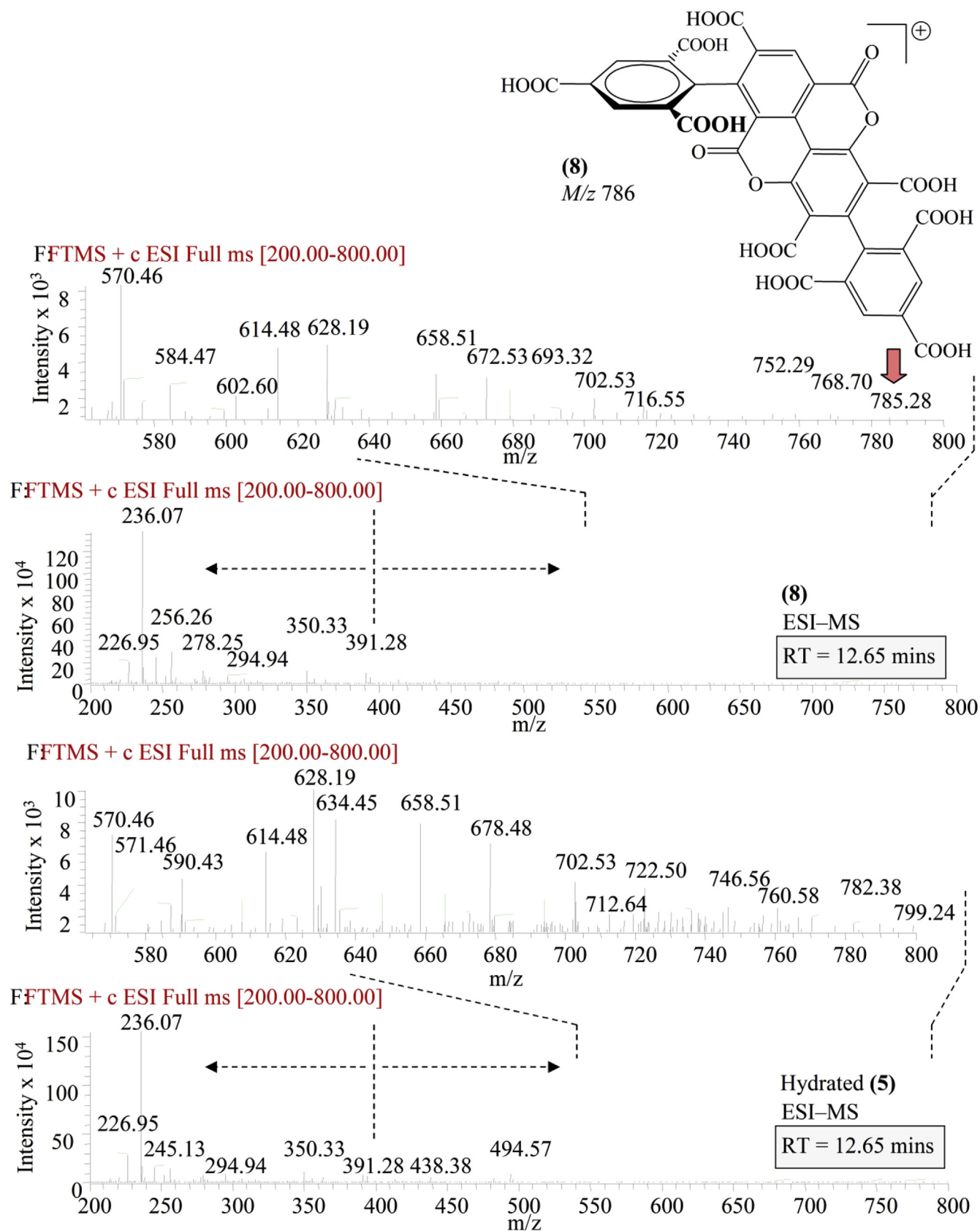


Fig. 5. Experimental ESI-MS spectra of (5) and (8) at RT = 12.65; Chemical diagram of the fragment ion at *m/z* 785 (Continue as Fig. S6).

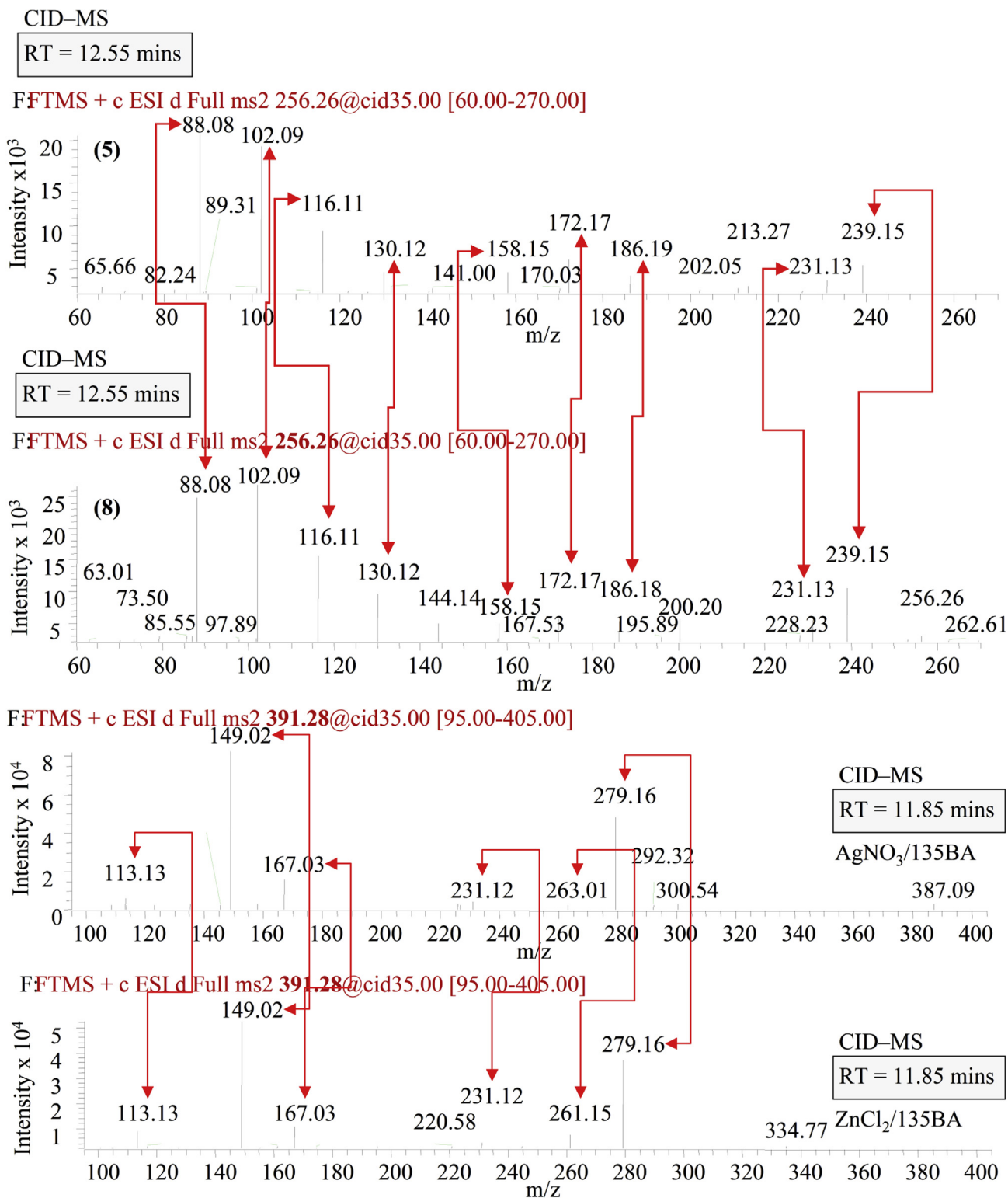


Fig. 6. CID-MS reactions of organosilver(I) and organozinc(II) catalysed products (5) and (8) (RT = 12.55 and 11.85) of ions at m/z 256 and 391 (Continue as Fig. S6); Correlation between MS fragment ions (Shown using arrows).

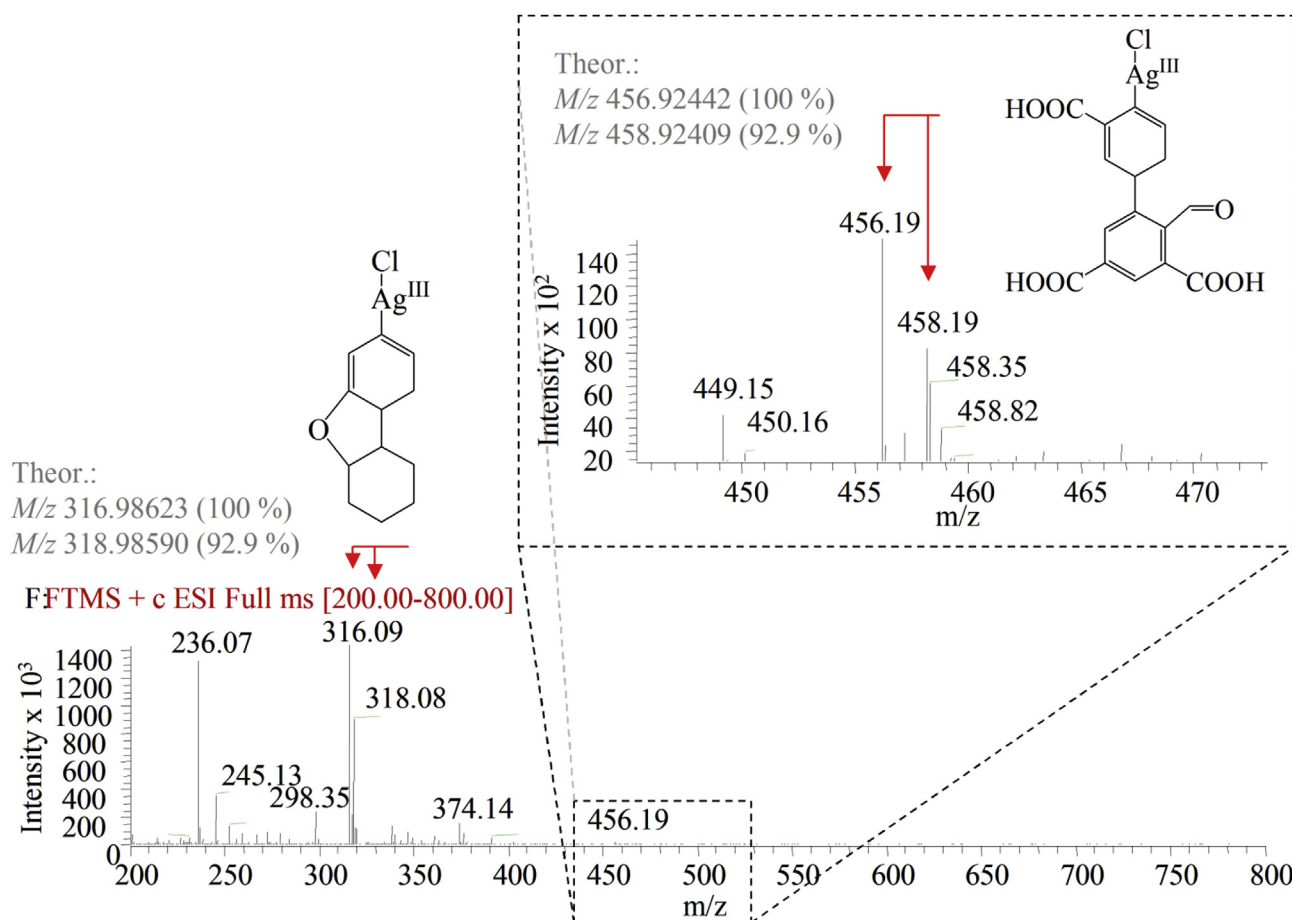


Fig. 7. ESI–MS spectra of Ag^{III} –organosilver intermediate detected in system 124BA/Ag at RT = 11.45 min (See Fig. 1); Chemical diagrams of the proposed organometallic fragment MS ions; Theoretical MS peaks of the isotopic shape of the Ag^{III} –organosilver intermediate.

at m/z 391 are depicted in Fig. 6 as well. The correspondence between the following rate constants of the fragment MS reactions are taken into consideration:

Ag(I)	CID–MS M/z 391 RT = 11.85 mins	Zn(II)
k_{Ag}^1		k_{Zn}^1
k_{Ag}^2	M/z 225.77	M/z 220.58 k_{Zn}^2
k_{Ag}^3	M/z 231.12	M/z 231.12 k_{Zn}^3
k_{Ag}^4	M/z 263.01	M/z 261.51 k_{Zn}^4
	M/z 279.16	M/z 279.16

The group tendency seen in the statistical Fig. 7 between the rate constants of the Ag(I) and Zn(II) catalysed reactions of 135BA (RT = 11.85) shown as k_{Ag}^i and k_{Zn}^i rate constants of i th reaction ($i = 1-4$) confirms the clear quantitative relationship between the reaction rates within one and the same type, but independent,

fragment CID reactions. A confidence level 90.51% is obtained. This result appears to corroborate the validity of the Iribarne-Thomson and Eyring model equation to CID–MS reaction kinetics, furthermore in statistically significant quantitative terms. The corresponding k_{Ag}^i and k_{Zn}^i values are: $k_{\text{Ag}}^1 = 0.00180641$, $k_{\text{Ag}}^2 = 0.001094573$, $k_{\text{Ag}}^3 = -2.577236 \cdot 10^{-4}$, $k_{\text{Ag}}^4 = 0.001007338338$, $k_{\text{Zn}}^1 = 0.001363357$, $k_{\text{Zn}}^2 = 0.001136825$, $k_{\text{Zn}}^3 = -4.09828 \cdot 10^{-4}$ and $k_{\text{Zn}}^4 = 0.00117052 \text{ s}^{-1}$, respectively. Turning to the concept of the thermodynamic and kinetic treatment of ESI–CID–MS reactions [12] it is shown in our current study its validity to quantitative correlational treatment of fragment MS reactions, thus leading to statistically significant correlation equations applicable within the framework of one and the same molecular scaffold of the analyte fragment molecules/ions under explicitly defined experimental conditions. As far as there are not many investigations which examine the quantitative kinetic and thermodynamic correlations between the rate constants of the CID–MS reactions and the energetics of the fragment ions, the new data reported in this paper provide excellent deal of evidence explaining the great reproducibility of CID–MS profile of the analyte in statistically significant quantitative terms. There is clear advantages of obtaining of the experimental statistical equations of the kinetic parameters of CID–MS reactions within the framework of the different classes of analytes in the way done here and in our previous contributions to this topic [12a], because of this approach allow unambiguous not only qualitative and quantitative analysis of the analytes but their 3D structural determination. The advantages include the fact that via this approach is extended significantly the

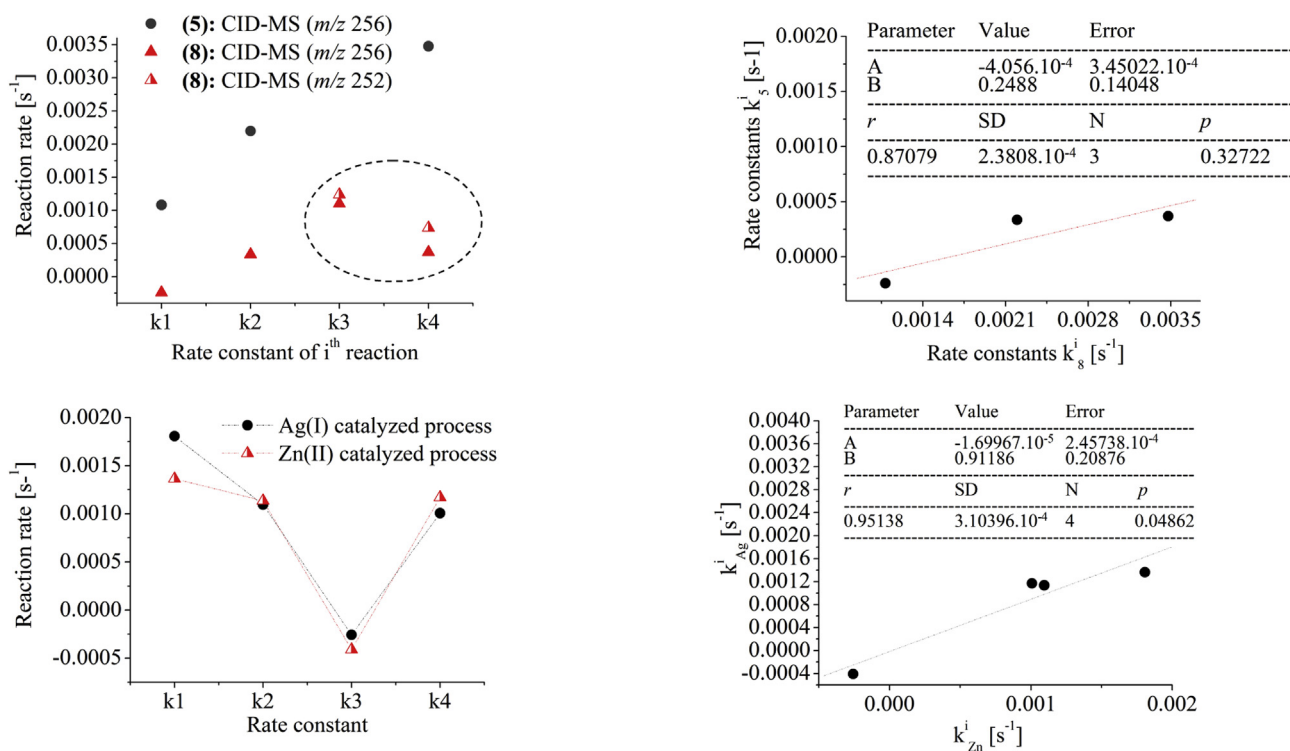


Fig. 8. Quantitative correlation of the experimental MS rate constants of the reactions; Chemometric analysis.

capability of CID–MS for structural 3D information, furthermore, as mentioned above in quantitative terms. Because of as it well known each 3D structure possesses a unique free Gibbs energy value, which within the framework of high resolution MS measurements can be experimentally obtained at very high accuracy. When the experimental MS data are correlated with high accuracy theoretical quantum chemical ones as demonstrated herein, there is gained a great power in the structural determination of analytes at very low concentrations. In this context our contributions in the sub-sections 3.2.2 and 3.3.1 are particularly relevant to research fields of reaction kinetics and catalysis; environmental analytical chemistry; agricultural chemistry; toxicology, etc. where an obtaining of high accurate analytical information at trace concentration levels is of paramount importance. As this brief comment has shown, it is very important to differentiate between the classical term ‘quantitative mass spectrometry’ dealing with quantitative determination of the analytes and ‘quantitative mass spectrometric reaction kinetics and thermodynamics’, dealing with the quantitative statistical relationships between rate constants and energetics of MS ions and their correlation to corresponding 3D molecular structures of the analytes and their conformations which is what this sub-section and corresponding Sub-section 3.3.1 and 3.2.1 set out to do.

4.3. Quantum chemical analysis

4.3.1. Mass spectrometric fragment reactions—kinetics and thermodynamics

There are various methodologies for computation of the reaction thermochemistry described in the literature. Arguably, the most well-worked out is the quantum chemical computation of the free Gibbs energy in solution using the polarizable continuum approaches (see the sub-section ‘Theory/computations’). Thus, the thermochemistry of the CID–MS fragment reactions discussed above is summarized in Tables S3 and S4. The quantitative

theoretical treatment of the thermodynamic parameters of the process $16e \rightarrow 16e^{1b}$ show $\Delta(\Delta G) = |0.2082|$ a.u. using the assumption about a heterolytic dissociation of the OH^- group from the ion at m/z 590 (The thermochemical data about the OH^- are $E^{ZVPE} = 6.49005$ kcal mol $^{-1}$ and $G = -74.530265$ a.u.). The corresponding computations about a homolytic dissociation process yielded to $\Delta(\Delta G) = |0.0795|$ a.u. when the thermochemical data of conformer 1 of the ion at m/z 573 is used (The thermochemistry of radical OH shows: 6.94875 kcal mol $^{-1}$, $G = -74.658573$ a.u.). In order to codify of our completed set of possible processes we were computed a hypothetical heterolytic dissociation reaction with a formation of H_3O^+ cation. The data show $\Delta(\Delta G) = |1.14|$ a.u. (The thermochemical data about H_3O^+ used in those computations are: 24.32617 kcal mol $^{-1}$, $G = -75.784106$ a.u.). As has been previously reported [12a] the complementary application of the experimental CID–MS thermochemical data and the corresponding 3D quantum chemical models of MS fragment ions allows to predict at a high level of reliability the exact molecular conformation and the 3D structure of the analyte, because of the methods of quantum chemistry, in particular, highlighting the computations of the free Gibbs energy parameter are highly sensitive towards any change of the electronic structure due to conformational perturbations, type hydrogen bonding network, environmental factors such as for example continuum polarity, ionic strength, T, P, etc. Despite the major utilization of CID mass spectrometry for quantitative analysis and 2D structural determination that are traditionally involved in quantitative analytical studies devoted to environmental chemical problematic, metabolomics, agrochemical problematics, etc., we argue with our current study that CID–MS has great capability for 3D structural analysis beyond it become routine application for quantitative analytical purposes. Due to the facts presented in this paper as well as those reported to [12a] more object oriented correlative studies between the theory and the experiment are needed to evidence the statement stated above. Cumulatively the

employment of the computational quantum chemistry into the treatment of experimental CID–MS reaction kinetic and thermodynamic data is of importance because of the high complexity of the analytes causes for a large set of possible 3D conformations and electronic structural redistribution along with various intraionic hydrogen bonded networks. Using MS measures for quantitative reaction kinetics and thermodynamics we are faced with the questions what molecular 3D conformation should be used as a model for quantum chemical computation of the theoretical parameters; where is the localization of the positive charge within the framework of the MS cation; where is the proton accepting centre in the molecular fragment, etc.? In addition, such as questions are not clear-cut within the framework of the different mass spectrometric methods, which are based on the different physical phenomena. In addressing problems such as these, mentioned above, we inevitably use a large set of possible 3D models in order to reproduce exactly the experimental reaction kinetics and thermodynamics. As it can be seen from the data in Tables S3 and S4 as well as Scheme S5, the free rotation around the COOH group in the cation at m/z 590 causes for a change of its free Gibbs energy $\Delta(\Delta G) = |0.00536|$ a.u. Or if we assume that this conformer (conformer 2) participates in the fragment homolytic reaction $1^6e \rightarrow 1^6e^{1b}$, than the $\Delta(\Delta G)$ value comparing the experimental ΔG (sub-section 3.2.2) and the theoretical one is $|0.06621|$ a.u. The corresponding quantity using the parameters of conformer 3 is $\Delta(\Delta G) = |0.074355|$ a.u. This result clearly shows the 3D conformer 2 of the MS ion at m/z 590 are participant in the fragment process. If we should establish a certain standard deviation of the $\Delta(\Delta G)$ between the theory and the experimental thermochemistry of the CID–MS reactions, we then should note point out that both high-to-ultrahigh resolution MS spectrometry and the high accuracy quantum chemical methods provide exact information about the energetics. In addition, it would seem that $\Delta(\Delta G)$ value is also affected by the initial charge localization within the framework of the molecular 3D model, at least indirectly, if we deduce the charge localization position on the base on the hypothesis that it should be localized within the framework of the region of the molecular residue where the bond cleavage is obtained. Because of if we hypothesise a random charge localization, for example, studying the molecular models of the charged ions at m/z 590 (Models I and II) the difference of the $\Delta(\Delta G)$ values show a clear influence of $\Delta(\Delta G) = |143.38|$ a.u. not only with the energetics, but also with the 3D molecular conformation. As it can be seen, however, the form (II) ($G = -2106.4378395$ a.u.) is not a resulting electronic distribution following the fragment pathways based on the cleavage of OH and COOH functional groups. The latter ones, however, are experimentally observed (Figs. 1–4). Tables S3 and S4 therefore show that depending on the design of a given molecular MS ion, theoretical free Gibbs energy value varies. Further research in this context will have to determine the boundaries of this variation within the framework of concrete class of molecular scaffolds of analytes; conformational changes; type of the substituents, etc., molecular and environmental factors. However, it is important to underline that the ΔG variation from class to class of molecular scaffolds allows to distinguish unambiguously from different hypothetical 3D structures of the analyte MS ions.

As far as there has been obtained a variation of the internal energy of the deuterium labelled benzoic acids [11e] our theoretical analysis includes computations of the thermochemistry of both labelled off and different deuterated derivatives. In the light of the known comprehensive knowledge about the deuterioexchange reactions of benzoic acids [11] it is important to underline that this molecular template in general appears straightforward one because of: (a) the initial fragmental reaction is without primary isotope effect, while (b) product of MS reaction involves loss of only one H

and/or D atoms [11e]. In this context the theoretical computational data in Table S3 of deuterium labelled fragment ions should be treated as possible independent products of MS fragment reactions instead of combination of corresponding MS fragment ions, which significantly simplifies the theoretical correlative analysis comparing with the experimental MS data shown in section 3.2. Importantly, however, the deuterium exchange contributes differently to the E^{ZPVE} and free Gibbs energy values showing differences $\Delta(\Delta G) = |2.10^{-5}|$ eV and $\Delta E^{ZPVE} = |0.11|$ eV, respectively. The data in Table S3 agree satisfactory well with the previous report on the deuterium labelled benzoic acid [11e] showing that the energetics of the relative loss of the OH/OD within the framework of the typical reaction as function of the internal energy is at about 12.7 eV [11e] studying the intensity ratio of the MS peaks at m/z 105/106 ($[M - OH]/[M - OD]$). A value 16 eV has been obtained studying the reaction of decarboxylation (MS intensity ratio of the peaks $[M - COOH]/[M - COOD]$). The MS intensity ratio $[M - OH]/[M - OD]$ is within 0.67–2.9, while the corresponding one $[M - COOH]/[M - COOD] > 20$ eV [11e]. It is important to underline that the data in Table S3 have not a representativeness from a statistical point of view because of we do not explicitly present a representative set of computations of off-labelled and deuterated derivatives in order to determine the range of ΔE^{ZPVE} and $\Delta(\Delta G)$ values for isotope labelled quaterphenyls.

4.3.2. Organosilver(I) catalysing reactions – mechanistic aspects

As essential part of the mechanism of organosilver catalysed C–H activation and C–C bond formation, it is important to discuss the formation of the derivative (2) in more detail. The mechanistic aspects of the reactions yielding to the products (2) and (3) can be described within the framework of the interactions of biphenyl-2,3,5,2',4',5'-hexacarboxylic acid (reagent R^1) and the corresponding silver ion containing organometallics 6'-substituted($\{AgCl\}$)-biphenyl-2,3,5,3',4'-pentacarboxylic acid (reagent R^2) and 5'-substituted($\{AgCl\}$)-biphenyl-2,3,5,2',4'-pentacarboxylic acid (reagent R^3) which are obtained via decarboxylation of R^1 and *o*- and *p*-positions (Scheme S4). The NBO charges are depicted in the same scheme. As can be seen most favourable for electrophilic reactions are the 6 and 6' positions of R^1 exhibiting $q_C(NBO) = -0.136$ and -0.170 , respectively. This result explains the preferred formation of MS fragment species of type 1^6e and 1^4g as MS ions (Figs. 1 and 2). On the other hand these data agree well with the proposed mechanism of formation of quaterphenyls (Scheme S4). The obtained charges $q_C(NBO) = -0.131$ (2'-position) and -0.127 (4'-position) explain the formation of the organometallic intermediates R^3 and R^4 via a decarboxylation reactions, however due to steric effects from the $-COOH$ groups at 3- and 5-position in R^1 there is unable formation of quaterphenyls with participation of the intermediate R^4 . On the contrary the products of interaction is explain exactly with intermediate R^3 . The steric effects is involved as reason preventing the formation of organosilver(I) intermediates at 3-, 6- and 6'-positions nevertheless that $q_C(NBO)$ values are -0.136 , and -0.170 , respectively. In this context we describe only R^5 as organometallic intermediate participating in the oligomer synthesis due to $q_C(NBO) = -0.126$ at 4-position in R^1 . The interaction of the organometallic intermediates R^2 and R^3 with R^1 at 3-position yields to the products $[1,1'; 2',3'; 1'',1''']$ quaterphenyl-2,3,5,4',5',2'',4'',5''',2''',3''',5'''-undecacarboxylic acid (5) and $[1,1'; 3',3'; 1'',1''']$ quaterphenyl-2,3,5,2',4',5',4'',6'',2''',3''',5'''-undecacarboxylic acid (6), respectively (Scheme S4). The experimental MS data about the new quaterphenyl derivatives of the polysubstituted benzoic acids foreground the reliability of the detail mechanism of C–C bond formation which has been described studying the biphenyl formation in Ref. [1c].

5. Conclusion

In the present paper our goals have been twofold. First we have considered the organosilver(I) catalysed reactions of synthesis and isolation of substituted quaterphenyls starting from the simple polysubstituted benzoic acids. In our contribution we have consistently attempted to draw the readers' attention on the insights of in collision induced mass spectrometric reaction kinetic and thermodynamics of the C–H activation and C–C/C–X (X = C or O) bond formation which has made possible to present the quantitative correlation between the intermolecular and environmental strongly impacting the reaction kinetics and thermodynamics. The experimental mass spectrometric kinetic and energy parameters are treated correlatively using the powerful approaches of the computational quantum chemistry as well as, thus providing a meaningful information about the correlation mass spectrometric reaction kinetic – thermodynamics and the corresponding molecular conformation and interaction networks of the analytes based on a comprehensive computational modelling. In this context the research beyond doubt contributes to the field of organometallic synthesis and catalysis providing a sound background for further effort to optimize the synthetic roads in order to gain an improvement of the reaction efficiency. Second, the paper provides quantitative treatment of the reaction kinetic and thermodynamic parameters associated with the concrete class of organic molecules, i.e. polysubstituted benzoic acids, which further has been correlated with the energetics of the corresponding 3D molecular conformation of the analyte thus significantly extending the capability of CID–MS to obtain exact 3D structure of the molecules on the base on the experimental collision induced reaction kinetics and thermodynamics of fragment ion reactions. In this context he presented excellent, statistically significant, correlational relationships call for the need for further methodological research effort on the complementary application of CID–MS and computational quantum chemistry of different classes of molecules. Because of the data reported require scrutiny as far as there is a clear statistical correlation between the experimental MS kinetic parameters within the framework of a concrete type of fragment reactions typical for given class of compounds as well as another set of statistical correlational equations typical of individual analytes, thus allowing to distinguish structurally very similar derivatives, furthermore in quantitative terms, despite the high complexity of the analytes as basic molecular scaffold as has been shown in this paper where the object of analyses are characterized with a significant complexity and high order similarity of the major molecular skeleton. The reliability of the statistical model equations correlating the reaction kinetics of the independent Zn(II) and Ag(I) catalysed processes but within the framework of one and the same fragment reaction types shows confidence level 90.51%. Our findings presented herein especially have an utmost important contribution in the exploration of CID–MS mass spectrometry for detail 3D structural elucidation in quantitative kinetic and thermodynamic terms allowing unambiguous differentiation of very complex from a molecular structural point of view analytes which is parallel are characterized with small differences of the chemical functional substitutions and/or change/perturbation of the 3D molecular conformation and/or intra/inter hydrogen bonding network on the base on the molecular energy parameters expressed in terms of free Gibbs energy and corresponding reaction kinetic parameters typical for selected set of fragment processes associated with the major molecular skeleton or concrete specific to the individual molecule substituent.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jorgchem.2017.09.035>.

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